Master Thesis

Investigation of neutral pion decay within the Dyson-Schwinger framework

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1 Introduction

We are now entering a time of precision particle physics. With the discovery of the Higgs boson in 2012 the final piece in the Standard Model has been experimentally verified. At the same time we are facing many unanswered questions: neutrino masses, dark matter, baryogenesis.

To gain further understanding new experimental data are needed. Present and future experiments can be roughly divided into two categories: high energy and high precision. The main example of the first type is the Large Hadron Collider (LHC) at CERN. The LHC was upgraded and began taking 13 TeV data in 2015. As data analysis continues in the coming years there is still great potential for new discoveries.

On the other hand as the difficulty and cost of high energy experiments increases, high precision experiments become increasingly important. A current example in this category is the g-2 experiment under construction at Fermilab. This experiment aims to obtain information about physics beyond the Standard Model by measuring the muon's magnetic moment to unprecedented precision. There is currently tension between the best theoretical and experimental values. For this reason the theory community is working towards lowering the uncertainty on the Standard Model prediction, where non-perturbative effects from QCD appear to dominate the error space. If the tension becomes stronger with improved precision on both sides, this would provide strong evidence for new physics.

Because of high precision experiments like g-2, improving predictions for decay amplitudes is becoming more important than ever. In this thesis we investigate the decays of the neutral pion. This particle is a meson which feels the strong force, as described by the underlying theory of Quantum-Chromodynamics (QCD). As the lightest QCD meson the pion can only decay into photons (γ) and leptons (e^+, e^-). The most common decay channel by far is the $\pi^0 \to \gamma\gamma$ process. This process is famously predicted by the QCD anomaly to fairly high precision. However this result assumes that the pion is massless in the decay amplitude and thus receives corrections of the order M_π^2/Λ_{QCD}^2, where M_π is the pion mass and Λ_{QCD} is the characteristic energy scale of the strong force. One of the aims of this work is to include these corrections to the dominant decay channel. Going beyond this we are also interested in various other decays of the neutral pion. In particular the three body Dalitz decay ($\pi^0 \to \gamma e^+ e^-$) and the ($\pi^0 \to e^+ e^-$) decay.

All of these decay processes can be determined using the ($\pi^0 \to \gamma\gamma$) form factor. This quantity must be determined non-perturbatively within QCD. This issue arises because the strong coupling constant becomes large at low energies. For this reason expansion in this constant fails for describing such low energy quantities. Non-perturbative methods overcome this problem by avoiding such expansions and working directly with the original mathematical formulation. In QCD such non-perturbative approaches can only be implemented numerically. Two methods in this field are numerical lattice QCD and the approach pursued in this work, the Dyson-Schwinger formalism. In lattice QCD a hyper cubic grid or lattice is used to define a discretized version of QCD that can be simulated on a (super) computer. By calculating with many different lattice spacings and extrapolating one aims to recover predictions of the continuum theory.

In the Dyson-Schwinger formalism the QCD correlators are related by an infinite set of coupled integral equations. These coupled equations are derived in a rigorous model independent way and contain a full description of all physical measurable processes. However the complicated form of these equations makes a direct solution, either analytic or numeric, impossible. For this reason various approximations are applied to reduce the equations to a set that can be solved numerically. In this thesis we will use the Dyson-Schwinger equations to perform the non perturbative calculation of neutral pion decays.

The two body ($\pi^0 \to e^+e^-$) decay is one of the least probable decays of the neutral pion. With a branching ratio of $BR(\pi^0 \to e^+e^-) = 6.64 \times 10^{-8}$ it is hard to measure and up until now experimental and theoretical results show a discrepancy up to 3 σ. This could be an indication of new physics and calls for new theoretical predictions as well as more experimental data. Within in the Dyson-Schwinger formalism it will be looked at for the first time.

This thesis is structures as follows. In the first section the basic theoretical structure behind the Dyson-Schwinger formalism is introduced, starting with general ideas in quantum field theory and working toward bound-state and the Dyson-Schwinger equations. The next section includes all ingredients needed for the calculation of the $\pi^0 \to \gamma\gamma$ form factor. This includes the basic calculation and structure of the quark propagator and the pion Bethe-Saltpeter amplitude and concludes in the expression for the form factor. The next section give a overview over the calculated results for the form factor. Followed by the discussion of the decay rate for other pion decays, namely the $\pi^0 \to \gamma e^+e^-$ Dalitz decay and the $\pi^0 \to e^+e^-$ decay.

2 Theoretical introduction

2.1 From Greens function to S-Matrix

The Greens functions are the connection between theoretical formalism and physical observables. They encode all the important physics and can be obtained using different approaches. In canonical quantization, fields of particles $\hat{\phi}(x)$ are treated as operators in a Hilbert space. These operators obey canonical (anti-) commutation relations and can be thought of in a particle anti-particle interpretation. In this approach the n-point Greens function is defined as the time ordered vacuum expectation value of products of fields:

$$G^{(n)}[\phi] = \left\langle \Omega | \mathrm{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\dots\hat{\phi}(x_n)\}|\Omega \right\rangle . \tag{2.1}$$

T indicates the time-ordered product and $|\Omega\rangle$ is the vacuum state. The time-ordered product is defined as

$$\langle\Omega|\mathrm{T}\hat{\phi}(x)\hat{\phi}(y)|\Omega\rangle \equiv \begin{cases} \langle\Omega|\hat{\phi}(x)\hat{\phi}(y)|\Omega\rangle, & \text{for } x^0 > y^0 \\ \langle\Omega|\hat{\phi}(y)\hat{\phi}(x)|\Omega\rangle, & \text{for } y^0 > x^0 \end{cases} , \tag{2.2}$$

for two fields, with the natural generalization for n fields. In short, it orders the operator in time, with the earliest time coordinate on the far right and the latest on the far left.

A different approach to define a Greens function is the path integral formalism. In this approach fields $\phi(x_i)$ are treated as functions instead of operators. In the path integral formalism the Greens functions are given by

$$G^{(n)}[\phi] = \frac{\int \mathcal{D}\phi\ \phi(x_1)\dots\phi(x_n)e^{-S[\phi]}}{\int \mathcal{D}\phi\ e^{-S[\phi]}} . \tag{2.3}$$

This is meant to integrate over all possible field configurations $\phi(\vec{x}, t)$ weighted by a phase factor that depends on the classical action S evaluated at the field configuration. The time ordering we explicitly need in the canonical definition is already incorporated into the definition of the path integral. In the integrand the fields are commuting numbers or in the case of fermions anti-commuting Grassmann variables. It is proven that equation (2.1) equals (2.3) and both sides are time-ordered.

The generating function of the full Greens function contains all information of the field theory and is analogous to the partition function in classical statistics. Therefore one way to fully define a quantum field theory is to write down its generating function, which is defined similar to the Greens function as

$$Z[J] = \int \mathcal{D}\phi e^{-S[\phi]+J\phi}. \tag{2.4}$$

Here ϕ stands for any field in the theory and J is the corresponding source to the field and $J\phi$ means $\int d^4x J_a(x)\phi_a(x)$. Greens functions are also obtained by taking the functional derivative of the generating function with respect to the sources J at zero external source.

$$G[\phi] = \langle\Omega|\, \mathrm{T}\, G[\hat{\phi}]\, |\Omega\rangle = \left[G\left[\frac{\delta}{\delta J}\right] \frac{Z[J]}{Z[0]} \right]_{J=0}. \tag{2.5}$$

This is usually denoted by $\langle\cdot\rangle$. The weighted functional averaged with the source set to zero. In the presence of an external source one writes $\langle\cdot\rangle_J$.
One distinguishes between the full, connected and one-particle-irreducible (1PI) Greens functions. The difference between these is easiest to visualize in terms of Feynman diagrams. While the full n-point Greens function contain all possible n-point diagrams, the connected ones include only those in which all end points are connected with at least one other point. The 1PI are all diagrams, where one internal line can be cut and the diagram stays connected.
The generating function Z generates the full Greens functions. One can also define a functional that generates only the connected Greens function which is related to Z through $W[J] = -\ln\ Z[J]$. The relation can be shown by a Taylor expansion. Furthermore we can do the same for the 1PI Greens function through Legendre transforming $W[J]$ into a functional called the effective action $\Gamma[\tilde{\phi}]$:

$$Z[J] = e^{-W[J]} = e^{(\Gamma[\tilde{\phi}] - \int_x \tilde{\phi} J(x))} \quad , \quad \Gamma[\tilde{\phi}] = W[J] - \tilde{\phi} J\ . \tag{2.6}$$

The averaged field $\tilde{\phi}$ is the vacuum expectation value of the field ϕ in the background of a source J. It can be expressed as

$$\tilde{\phi}(x) = -\frac{\delta W[J]}{\delta J(x)} = \frac{1}{Z[J]} \frac{\delta}{\delta J(x)} Z[J] = \langle \phi(x) \rangle\,. \tag{2.7}$$

As already mentioned above, we can create every state that we are interested in by acting with field operators on the vacuum. The Greens functions can then be understood as inner products of these states. For example a four-point Greens function can be understood as the inner product of a bra-state $\langle 0|\phi\phi$ with a ket state $\phi\phi|0\rangle$. Each of these can be decomposed in eigenstates of the hamiltonian, and generally we expect all states with the proper quantum numbers to appear in the decomposition. Thus the full Greens function includes inner products with all possible hamiltonian states that have the quantum numbers of the operators. In practice we are often only interested in one of these states and we therefore need a method that helps us extract it. The Lehmann-Symanzik-Zimmermann reduction formula (LSZ) achieves exactly this. One applies LSZ to the Greens function by amputating it and putting it on shell. The latter means that the time-components of momenta are evaluated at energies equal to the physical energy of the particle. This procedure picks out particular scattering states within $\langle 0|\phi\phi$ and $\phi\phi|0\rangle$. In the Greens function, we are thus left with an inner product of scattering states. This is the so called S-matrix. More precisely the S-matrix is an infinite dimension matrix whose entries are all possible inner products of scattering states.

In certain cases the S-matrix elements can be calculated perturbatively. To keep track of such perturbative expansions it is convenient to use Feynman diagrams. Each diagram corresponds to a mathematical expression and evaluating these can give a perturbative prediction for the S-matrix. For a given process one draws all the possible Feynman diagrams associated with the incoming and out-going particles in the game. The allowed vertices come directly from the Lagrangian of the theory and give the rules for drawing and evaluating the diagrams. The diagrams are grouped by their order in terms of some coupling constant and perturbative predictions are only possible if higher orders are suppressed.

An example where perturbation theory works well is quantum electrodynamics (QED), the quantum theory of Maxwell's equations. Here the coupling constant in the Lagrangian is the fine-structure constant α_{em}. This constant is much smaller than one and we can use it as an expansion parameter to calculate diagrams order by order in α_{em}. This leads to high precision predictions of electromagnetic processes.

The coupling constant of the strong force (QCD), by contrast, is unfortunately not a useful expansion parameter. It grows as energy decreases, which makes it as useful parameter in processes with high momentum transfer, but leading to a failure of perturbation theory for low energy processes. This means that higher order diagrams in an expansion in powers of α_S become more an more relevant and cannot be neglected. Thus using perturbation theory for strong interaction processes would naively require summing an infinite set of diagrams. This is not practically possible, and it would in fact fail even if it were, since the perturbative series is an asymptotic expansion.

This calls for the use of non-perturbative methods. One such method is the Dyson-Schwinger formalism, which will be used in course of this thesis.

2.2 QCD action

Generally a quantum field theories start with constructing a Lagrangian density, $\mathcal{L}$, or equivalently its action ($S = \int d^4x\mathcal{L}$). In this thesis we are interested in the decay of the neutral pion, the lightest particle that feels the strong force. Quantum Chromodynamics (QCD) is the theory of strong interactions and it is described by a local, non-Abelian $SU(3)$ gauge symmetry. The gauge invariant **QCD action** is given by

$$S_{QCD} = \int d^4x \left(\bar{\psi}(\not{D} + m)\psi + \frac{1}{4} F^a_{\mu\nu} F^a_{\mu\nu} \right) , \tag{2.8}$$

with the fermion fields ψ, the covariant derivative $D_\mu = \partial_\mu - igA_\mu$, the coupling constant g and the field-strength tensor $F_{\mu\nu}$. All equations and calculations in this thesis are given in Euclidean space. The covariant derivative ensures that the action is invariant under a local $SU(3)_{\mathrm{Color}}$ transformation $\psi' = U(x)\psi$. In order to define this new derivative we introduced a new field into the theory, namely the gluon field A_μ. This field is an element of the $SU(3)$ Lie algebra $A^\mu = \sum A^\mu_a(x)t_a$. The gluon field-strength tensor is defined as $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + ig[A_\mu, A_\nu]$, which is nothing different than the commutator between two covariant derivatives and is therefore also an element of the Lie algebra $F_{\mu\nu} = \sum F^a_{\mu\nu} t^a$. Under the $SU(3)_C$ transformation $U(x) = e^{i\sum_a \varepsilon_a(x)t_a}$ the fields transform as

$$\psi' = U\psi \ , \quad A'_\mu = UA_\mu U^\dagger - \frac{i}{g} U(\partial_\mu U^\dagger) \ , \quad F'_{\mu\nu} = UF_{\mu\nu}U^\dagger, \tag{2.9}$$

The matrices t_a used above are basis elements of the $SU(3)$ Lie algebra. There are 8 such elements in $SU(3)$ so that $a = 1, \cdots, 8$. In the fundamental representation the generators of $SU(3)$ are often expressed in terms of the Gell-Mann matrices, denoted by λ_a, with $t_a = \lambda_a/2$. With this definition of the QCD action, the Greens functions form the former section are fully defined.

2.3 Dyson-Schwinger equation

In the early fifties Dyson developed an equation for summing up an infinite set of Feynman diagrams [1, 2]. Later, Schwinger encountered the same set of equations and extended the work Dyson had begun to n-point Greens function [3]. Thus the equations are called the Dyson-Schwinger equations or also the quantum equations of motion, since they are similar to the Euler-Lagrange equations in classical mechanics.

In classical field theory the goal is to determine the field configurations ϕ for which the action is stationary. These are the configurations that satisfy

$$\frac{\delta S[\phi]}{\delta \phi} = 0 \, . \tag{2.10}$$

For example applying eq. (2.10) to the electromagnetic action gives Maxwell's equations, and solving these gives the electric and magnetic fields of a given classical system.

To connect this to quantum field theory we first revisit the generating functional $Z[J]$, defined in eq. (2.4), which we repeat here for convenience

$$Z[J] = \int \mathcal{D}\phi e^{-S[\phi]+J\phi} \,. \tag{2.11}$$

Note that this integral, like all integrals, must be invariant under a change of integration variable $\phi \to \phi'$, as long as the change is performed properly keeping track of the effect on the integrand and the measure. A special choice here is the infinitesimal change of variables $\phi \to \phi + \delta\phi$. This change leaves the measure invariant $\mathcal{D}(\phi+\delta\phi) = \mathcal{D}\phi$ and we deduce

$$\int \mathcal{D}\phi e^{-S[\phi]+J\phi} = \int \mathcal{D}\phi e^{-S[\phi+\delta\phi]+J(\phi+\delta\phi)} \,. \tag{2.12}$$

Expanding the right-hand side, and requiring the $\mathcal{O}(\delta\phi)$ term to vanish, we deduce

$$\left\langle \frac{\delta S[\phi]}{\delta\phi(x)} \right\rangle_J = J(x) \,, \tag{2.13}$$

where the subscript indicates that the source J is not set to zero, i.e. for any function $f(\phi)$,

$$\langle f(\phi) \rangle_J \equiv Z[J]^{-1} \int \mathcal{D}\phi f(\phi) e^{-S[\phi]+J\phi} = Z[J]^{-1} f\left(\frac{\delta}{\delta J}\right) Z[J] \,. \tag{2.14}$$

Combining these results we deduce

$$\left\langle \frac{\delta S[\phi]}{\delta\phi(x)} \right\rangle = Z[J]^{-1} \frac{\delta S}{\delta\phi} \left[\frac{\delta}{\delta J(x)}\right] Z[J] = J(x), \tag{2.15}$$

which means that the quantum average of the functional derivative of the action in presence of a source equals the source. If we replace the generating functional Z with the effective action Γ introduced in eq .(2.6), eq. (2.15) becomes:

$$\frac{\delta\Gamma}{\delta\tilde{\phi}(x)} = \frac{\delta S}{\delta\phi} \left[\tilde{\phi}(x) + \int_y \frac{\delta^2\Gamma}{\delta\tilde{\phi}(x)\delta\tilde{\phi}(y)} \frac{\delta}{\delta\tilde{\phi}(y)}\right] . \tag{2.16}$$

More detail on the derivation that lead to eq. (2.14) and eq. (2.16) can be found in the appendix (7.3.1). Eq. (2.16) is the generating Dyson-Schwinger equation (DSE) for the 1PI Greens function. With this powerful equation we can derive the DSE for the quark propagator, or any other field existing in the theory. This is achieved by taking derivatives of the effective action with respect to the field and setting the averaged field $\tilde{\phi} = 0$. The Dyson-Schwinger equation for the quark propagator is given by:

$$S^{-1} = S_0^{-1} + \int g\gamma^\mu D^{\mu\nu} S \,\Gamma^\nu, \tag{2.17}$$

with the fermion propagator S, the bare propagator S_0, the bare fermion-gauge-boson vertex γ^μ, the dressed vertex Γ^ν and the gauge boson propagator $D^{\mu\nu}$. Here we suppressed all arguments for brevity, the DSE will be discussed furthermore in section (3.1). In QCD the gauge-bosons are called gluons. For more detail on the derivation of eq. (2.17) the reader is also referred to the appendix (7.3.2). Note that the S in eq. (2.17) is not the action as used in the equations above but the quark propagator S.

Eq. (2.17) is an iterative equation and results in a coupled set of integral equations. If we were able to solve this system, we would have solved the theory and we would know all possible observables. The quark-gluon vertex and the gluon propagator as parts of the quark DSE, fulfill their own DSE which needs the knowledge of higher Greens functions, such as the triple gluon vertex and so on. Together this combines into an infinite tower of coupled integral equations, which is impossible to solve in general. We need to make a truncation on the system to make it solvable. This truncation needs to respect the symmetries of the theory in order to reduce the effect on physical observables. The truncation used in the majority of DSE studies is the rainbow-ladder truncation, which will be introduced in a following section.

2.4 Bound-state equation

The n-point Greens function can be related to a n-point interaction matrix $\mathcal{T}^{(n)}$, which includes all possible interactions between n particles. We can rewrite the n-point Greens functions in terms of this $\mathcal{T}$-matrix and a non-interaction term G_0:

$$G^{(n)} = G_0^{(n)} + G_0^{(n)} \mathcal{T}^{(n)} G_0^{(n)}. \tag{2.18}$$

G_0 is a product of n fully dressed quark propagators, as diagrammatically indicated for the case of $n = 4$ in the upper right corner of Fig.(2.18). The $\mathcal{T}$-matrix contains all amputated and connected n-point diagrams with n incoming and n outgoing particle. Amputated means that the diagrams have no external legs and this is indicated by the dotted lines in Fig.(2.18). All dirac, flavor and color indices are suppressed for the following equations, but will be shown explicitly in the case for the meson in section (3.2). The n-quark $\mathcal{T}$-matrix can be further related to a n-quark scattering kernel $\mathcal{K}^{(n)}$, lower corner left side Fig.(2.18):

$$\mathcal{T}^{(n)} = \mathcal{K}^{(n)} + \mathcal{K}^{(n)} G_0^{(n)} \mathcal{K}^{(n)} + \mathcal{K}^{(n)} G_0^{(n)} \mathcal{K}^{(n)} G_0^{(n)} \mathcal{K}^{(n)} \dots, \tag{2.19}$$

which can be rewritten such that

$$\mathcal{T}^{(n)} = \mathcal{K}^{(n)} + \mathcal{K}^{(n)} G_0^{(n)} \mathcal{T}^{(n)}. \tag{2.20}$$

This is an iterative equation and it is exact. Terms up to any order are included since the interaction kernel is iteratively inserted on both sides. Later on we will see that in the

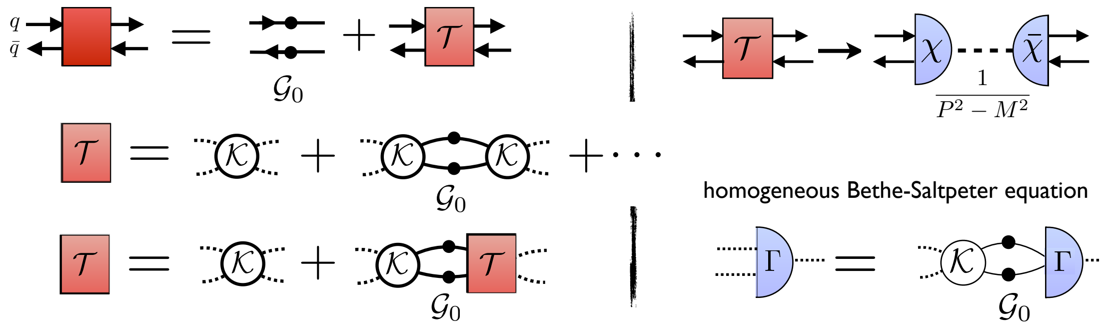


Figure 2.1: The bound-state equation expressed in figures. The doted lines indicate that the propagators lines are amputated. The left side shows the decomposition of the full n-point (in this case 4-point = 4 external quark lines) Greens function. The right side shows the approximation for the bound-state amplitude close to a hadron pole and the bound state equation in case of a 4-point function: the homogenous Bethe-Salpeter equation.

course of our bound-state calculation a finite expansion of the kernel is used. Eq. (2.18) can be rearranged into

$$\left(G^{(n)}\right)^{-1} = \left(G_0^{(n)}\right)^{-1} + \mathcal{K}^{(n)}. \tag{2.21}$$

At low energies, quarks are confined and $\mathcal{T}$ does not describe scattering, but we can calculate another quantity with it, hadron masses. It is known that bound states appear as poles in QCD Green functions. This insight is easily realized using the Källen-Lehmann spectral representation for the propagator and the completeness relation of the QCD Fock space, to rewrite the four-point function.

As an example the derivation for the propagator (2-point Greens function) can be found in the appendix (7.3.4). For mesons (quark-antiquark) we investigate the 4-point Greens function $G^{(4)}$, for baryons the 6-point functions $G^{(6)}$. The 4-point Greens function for example is given by

$$G^{(4)}_{\alpha\beta\gamma\delta}(x_1, x_2, x_3, x_4) = \langle 0|\mathrm{T}\psi_\alpha(x_1)\overline{\psi_\beta}(x_2)\psi_\gamma(x_3)\overline{\psi_\delta}(x_4)|0\rangle \tag{2.22}$$

where T is again the time-ordering operator.

As mentioned above bound states appear as poles in n-point Greens functions. To research a bound-state we are in particular interested in the behavior of the n-point Greens function close to the pole. In this region the pole structure dominates and we can approximate the n-point function by a transition element $\chi^{(n)}$ and the explicit pole structure (upper left corner Fig(2.1)). Using the completeness relation the Greens function becomes

$$
\begin{aligned}
\int d^4x_2 e^{ix_2P} G^{(2)}_{\alpha\beta\gamma\delta}(x_1,x_2,x_3,x_4) \quad &\longrightarrow \quad \chi^a_{\alpha\beta}(x_1,P)\frac{\mathcal{N}}{P^2-M^2_\lambda}\bar{\chi}^a_{\gamma\delta}(x_3,P) \\
\chi^a_{\alpha\beta}(x_1,p) \quad &= \quad \langle 0|\mathrm{T}\psi_\alpha(x_1)\overline{\psi_\beta}(0)|\lambda_a(p)\rangle,
\end{aligned}
\tag{2.23}
$$

where the arrow indicates that we are looking close to the pole and where we have introduced the so called Bethe-Salpeter wave function (**BSW**) $\chi^a_{\alpha\beta}(x,p)$ of a hadron state $\lambda_a(p)$. $\mathcal{N}$ is a constant, which depends on the spin of the resulting particle. In case of a pseudo-scalar meson, like the pion, $\mathcal{N}=1$. The greek letters are Dirac indices. The BSW gives the overlap between the two-quark state, defined by the quark fields acting on the vacuum, and the hadron state, $\lambda_a(p)$. Thus the BSW is often referred to as transition element between the vacuum state and a hadron state with momentum p.

Applying the near pole expression to eq. (2.20) we end up with the homogeneous Bethe-Salpeter equation (**BSE**) for a bound state λ. This is given diagrammatically on the bottom left in Fig.(2.1) and mathematically by

$$
\Gamma^{(n)} = G_0^{(n)}\mathcal{K}^{(n)}\Gamma^{(n)}. \tag{2.24}
$$

$\Gamma^{(n)}$ is the the amputated BSW, which is called the Bethe-Salpter amplitude (**BSA**) for $n=4$. The BSW is related to the BSA via $\chi^{(n)}_\lambda = G_0^{(n)}\Gamma^{(n)}_\lambda$.

Normalization

Eq. (2.24) is only defined up to a multiplicative constant. A physically motivated normalization condition is thus needed. The canonical norm demands that the residue at the bound state pole equals one. Only in this case the decomposition in eq. (2.23) holds exactly. Let us consider an n-point Greens function close, or at the hadron pole with an arbitrary residue in r_λ:

$$
G^{(n)}(p,P) = r_\lambda\,\frac{\chi^{(n)}\ \mathcal{N}\ \bar{\chi}^{(n)}}{P^2-P^2_\lambda}, \tag{2.25}
$$

where $P^2_\lambda = M^2_\lambda$ equal the mass of the bound state λ. Using the relation introduced in eq. (2.21) we further consider the partial derivative of $G^{(n)}$ with respect to the total momentum P:

$$
\begin{aligned}
\frac{\partial}{\partial P_\mu}(G^{(n)})^{-1} \quad &= \quad (G^{(n)})^{-1}\left(\frac{\partial}{\partial P_\mu}G^{(n)}\right)(G^{(n)})^{-1} \\
\frac{\partial}{\partial P_\mu}G^{(n)} \quad &= \quad G^{(n)}\frac{\partial}{\partial P_\mu}\left((G_0^{(n)})^{-1}-\mathcal{K}^{(n)}\right)G^{(n)}.
\end{aligned}
\tag{2.26}
$$

Now we use the pole assumption eq. (2.23) and insert it for $G^{(n)}$ in eq. (2.26), which results in

$$\begin{aligned} 2\,\frac{P_\mu}{r_\lambda} &= \bar{\chi}^{(n)}\frac{\partial}{\partial P_\mu}\left(G_0^{(n)-1} - \mathcal{K}^{(n)}\right)\chi^{(n)} \\ &= \bar{\Gamma}^{(n)}\left(\frac{\partial}{\partial P_\mu}G_0^{(n)-1} - G_0^{(n)-}\left(\frac{\partial}{\partial P_\mu}\mathcal{K}^{(n)}\right)G_0^{(n)}\right)\Gamma^{(n)} \end{aligned} \tag{2.27}$$

For $r_\lambda = 1$ this is the so called **canonical normalization condition**. A normalization factor N is introduced to eq. (2.27) such that $(1/N^2 \cdot r_\lambda = 1)$. To get a unit residue we thus have to multiply the bound state amplitude by the root of the normalization factor $\Gamma^{(n)}\sqrt{N}$ and eq. (2.27) holds.

With this canonical normalization condition eq. (2.27) and the homogenous bound state equation eq. (2.24), the bound state wave function χ on the mass shell, or equivalently the bound state amplitude Γ, are completely determined.

2.4.1 Bethe-Salpeter equation

The homogenous Bethe-Salpeter equation (**BSE**) is the $n = 4$ case of the general bound state equation eq. (2.24). In this particular case the vertex amplitude has two independent momenta, the total momenta of the bound state P and the relative momentum between the two out-going quarks p. The $G^{(4)}$ Greens function was given in eq. (2.22). The homogenous BSE eq. (2.24) in the case of $n = 4$ reads as:

$$\left[\Gamma_\lambda^{ab}(p,P)\right] = \int_q \mathcal{K}_{ab}^{a'b'}(q,p,P)\,[S(q_+)\Gamma_\lambda(q,P)S(q_-)]^{a'b'}\,, \tag{2.28}$$

with the renormalized amputated $q\bar{q}$ interaction kernel $\mathcal{K}$, the fully dressed quark propagator S and the Bethe-Saltpeter amplitude Γ_λ (BSA) for a meson λ. The letters a, b indicate a collective index of flavor, color and Dirac indices. The incoming and out-going quark momenta are $q_+ = q+\eta P$, $q_- = q+(1-\eta)P$, with the loop momentum q, the total momentum of the bound state P and the momentum distribution parameter $\eta \in \{0, 1\}$. A graphical form of the BSE is given in the lower right corner of Fig. (2.1) and are more detailed version in Fig. (3.4). The interaction kernel includes an infinite set of diagrams describing the interaction between a quark and an anti-quark and is 2-PI in the s-channel. A diagrammatical expression of the skeleton expansion for the kernel is given in Fig.(2.5). All these objects are contracted and integrated over the loop momentum q, with the notation $\int_q = \int d^4q/(2\pi)^4$. The quark propagators are obtained by solving the DSE, as given in eq. (2.17). More detail on the calculation will follow in the next section.

To obtain the normalized BSA Γ the amplitude has to be rescaled to fulfill the canonical normalization condition eq. (2.27). In course of this thesis the total momentum

$$2\delta^{ij} P_\mu = \frac{d}{dP_\mu}$$

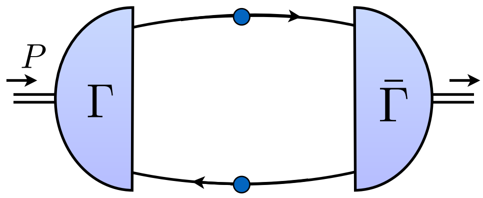


Figure 2.2: Canonical quantization condition for the Bethe-Salpter amplitude. With the Bethe-Salpter amplitude Γ and the complex conjugated amplitude $\bar{\Gamma}$. As given in eq. (2.30).

of the meson will be on-shell and at rest. In this case the normalization condition eq. (2.27) simplifies, because the kernel does not depend on the total momentum P, thus the derivative only acts on the propagators S:

$$2P_\mu = \bar{\Gamma} \frac{\partial}{\partial P_\mu} \left(S_+ S_-\right) \Gamma. \tag{2.29}$$

If we multiply with P_μ and spell out the integration over all possible relative momentum p the condition eq. (2.29) follows as

$$\delta^{ij} = \frac{P_\mu}{2P^2} \frac{\partial}{\partial P_\mu} \int_q Tr_{DFC} \left[\bar{\Gamma}^i(q,Q)\ S(q+\eta P)\ \Gamma^j(q,Q)\ S(q-\bar{\eta}P)\right]\Bigg|_{P^2=M_-\lambda^2}, \tag{2.30}$$

which can be seen diagrammatically in Fig.(2.2) and where $\bar{\Gamma}$ is the complex conjugated amplitude. The expression is evaluated at the mass of the bound state and $Q^2 = -M_\lambda^2$ is a fixed momentum set to the bound state mass. We can substitute the derivative such that $\partial/\partial P_\mu = 2P^\mu\ \partial/\partial P^2$, which will cancel the fraction and simplify eq. (2.30).

2.5 Renormalization

An effect that arises in quantum field theory is the need for regularization and renormalization. Some of the integrals in the calculation of Greens function or equivalently DSE ad BSE calculations will be divergent. The problem arises from the fact that the equations given above generally leads to infinities in perturbative calculations, and are not well defined non-perturbatively. To overcome this issue we put in a regulator. One example of this is a hard momentum cutoff on divergent integrals. But since this choice is arbitrary it should not affect physical predictions. This is where renormalization comes in. The basic idea is to choose the bare, fundamental parameters in the action in a way that produces consistent predictions. As the cutoff, or other regulator, is changed, these bare parameters need to changed as well in order to keep the right physical predictions. We will talk about this idea in more detail in section (3.1) below, where we show how it works in the specific context of the quark DSE calculation. A introduction to the general topic is given in appendix (7.3.5).

2.6 Symmetries

The symmetries of the QCD Lagrangian are an interesting tool to gain insight to relation between Greens functions. Through conservation and invariance of field we derive identities such as the Ward-Takahashi identities (**WTI**). The WTI's are symmetry relations between n-point and $n+1$-point Greens functions and they show the current conservations. The global transformations of our interest are $U(1)_v \times SU(N_f)_V \times SU(N_f)_A \times U(1)_A$ and the corresponding symmetries and currents.

The knowledge about conserved symmetries of the theory is also valuable when using approximations or truncations. As the DSE eq. (2.17) and equivalently the BSE are not directly solvable we need to make a truncation on our system. In this work we use the Rainbow-ladder truncation, which will be subject of the following section.

2.6.1 Current Algebra

We begin the discussion of symmetries with Noether's theorem. Noether's theorem states that every symmetry transformation which leaves the theory invariant implies the existence of a conserved current. The corresponding charge is a constant of motion. In other words the Noether charge commutes with the hamiltonian. This means that the eigenstates can be organized according to their quantum numbers under these charges. For example the $U(1)$ symmetry of QED implies the conservation of electric charge and thus all particles can be classified using the quantum number of this symmetry, e.g. the neutron has charge zero and the proton plus one.

The following table gives a quick overview of the important chiral symmetries of QCD, with their corresponding transformations. Considering QCD with three light quarks $N_f = 3$. The table also indicates if the Lagrangian is invariant under the transformation, and gives the corresponding current and the charge if it exists.

transformation	condition for $\Delta L = 0$	current J_μ	charge
$U(1)_V : e^{i\varepsilon}$		$\partial_\mu J^\mu = 0$, $J_\mu = \bar{\psi}\gamma_\mu\psi$	bayron number
$U(1)_A : e^{i\gamma_5\varepsilon}$	if $m = 0$	$\partial^\mu J^5_\mu = 2i\bar{\psi}m\gamma_5\psi$ $-Tr[\mathbb{I}_F]\frac{g^2}{32\pi^2}\varepsilon^{\alpha\beta\mu\nu}F_{\alpha\beta}F_{\mu\nu}$ + e.m anomaly	
$SU(3)_V : e^{i\sum_a \varepsilon^a\lambda^a}$	if $m_u = m_d = m_s$	$\partial^\mu J^a_\mu = i\bar{\psi}\left[\lambda^a, M\right]\psi$	Isospin & Hypercharge
$SU(3)_A : e^{i\sum_a \varepsilon^a\lambda^a\gamma_5}$	if $m = 0$	$\partial^\mu J^{a5}_\mu = i\bar{\psi}\{\lambda^a, M\}\psi$ $-\mathrm{Tr}_{CF}[Q^2\frac{\lambda^a}{2}]\frac{e^2}{16\pi^2}\varepsilon^{\alpha\beta\mu\nu}F_{\alpha\beta}F_{\mu\nu}$	

In the table M is the mass matrix that includes all quark masses on its diagonal, ψ is a fermion field, $F^{\mu\nu}$ is the field strength tensor and Q is the quark charge matrix. The generators t_a of the $SU(3)_f$ symmetry, are the Gell-Mann matrices λ_a. There are 8 generators for $SU(3)$. The subscript A indicates that the transformation leads to a conserved axial-vector current and V indicates a vector current.

The $U(1)_A$ symmetry is broken explicitly by the mass m. Even if the mass goes to zero, the symmetry is anomalously broken due to the second term, which is the so called $U(1)$-anomaly. In this case the anomaly contribution depends on the number of flavors. $SU(3)_V$ is an approximate symmetry of QCD. It holds for the approximation that all quark masses are equal, which is a reasonable approximation for up and down quark but the strange quark is much heavier. The $SU(3)_A$ symmetry is conserved in the limit of zero quark mass $m = 0$ and is explicit broken for a non-zero mass. Besides the QCD term, $SU(3)_A$ also contains a anomaly term, which come from electro-magnetic effects (second term in the $SU(3)$ column). This term arises because quantum corrections break the symmetry even for vanishing quark masses. The $SU(3)$ flavor axial anomaly plays an important role in the prediction for the neutral pion decay ($\pi^0 \to \gamma\gamma$) as we will see later on.

The currents of the symmetries given in the table can be constructed using quark fields and different tensor structures in form of Dirac matrices. We can construct the following quark bilinears

$$j_a^{\Gamma}(x)_{(R,L)} := \bar{\psi}(x)_{R,L}\Gamma\ t_a\psi(x)_{R,L} \quad , \quad \Gamma \in \{\gamma^\mu, \gamma^\mu\gamma_5, \mathbf{1}, i\gamma_5\}. \tag{2.31}$$

The different choices for Γ correspong to the different currents $j_a^{\Gamma}(x)$ (vector, axial-vector, pseudo-scalar and scalar). The subscripts R and L on the quark fields indicate whether they are right- or left-handed fields. These are defined via projection operators as

$$\psi_R(x) \equiv \frac{1+\gamma_5}{2}\psi(x)\,, \qquad \psi_L(x) \equiv \frac{1-\gamma_5}{2}\psi(x)\,. \tag{2.32}$$

Linear combinations of the bilinears construct the vector and axial-vector current: $V_a^\mu(x) = (j_a^{\gamma^\mu})_R + (j_a^{\gamma^\mu})_L$ and $A_a^\mu(x) = (j_a^{\gamma^\mu\gamma_5})_R + (j_a^{\gamma^\mu\gamma_5})_L$.

The charge corresponding to a current is the spatial integration over the current $Q_a^{\Gamma} = \int d^3x\ j_a^{\Gamma}$. In the case that the current is a four-vector, the time component is used in to define the charge. Through the charge we can define the so called current algebra, which gives the commutator relation between charges or currents. This algebra is part of the derivation for the Ward-Takahashi identities and for a more detailed discussion the reader is encouraged to check [4].

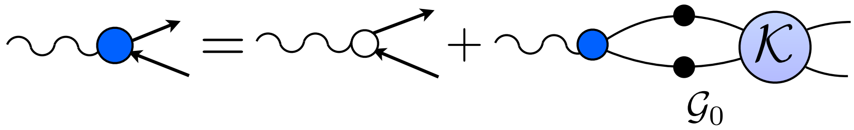
A interesting limit is the chiral limit, where all the quark masses are zero, $m = 0$. In this limit the Lagrangian is invariant under $SU(N_f)_A \times SU(N_f)_V \simeq SU(N_f)_L \times SU(N_f)_R$. This flavor symmetry is an approximate symmetry called the chiral symmetry. We can derive chiral limit relations between different dressing functions of the quark and pion, which will be discussed in the next section.

2.6.2 Ward-Takahashi identities and the Gell-Mann-Oakes-Renner relation

The keystone for understanding dynamical-chiral symmetry breaking (**DχSB**) and establishing the Ward-Takahashi axial-vector identities (**AVWTI**) is the color-singlet axial-vector vertex $\Gamma_{5\mu}$, which can be seen in Fig.(2.3). $\Gamma_{5\mu}$ is the solution to an inhomogeneous Bethe-Salpter equation and can be expressed in terms of a three-point Greens function:

$$G^{\Gamma}_{a,\alpha\beta}(x_1, x_2, x_3) := \langle 0|\mathrm{T}j_a^{\Gamma}(x_1)\psi_\alpha(x_2)\bar{\psi}_\beta(x_3)|0\rangle,\ \Gamma = \gamma^5\gamma^\mu \tag{2.33}$$

with the axial-vector current given by $j_a^{5\mu}(x)$, given in eq. (2.31). Since the current corresponds to two quark fields $\psi(x)$ and $\overline{\psi}(x)$, the three-point function could also be seen as a four-point function with two fields having the same spatial coordinate x. If we



$$= \langle 0|\mathrm{T}\psi(x_1) j^{5\mu}(x)\overline{\psi}(x_2)|0\rangle$$

Figure 2.3: The axial-vector vertex inhomogenous BSE: $\sim\ \Gamma_{5\mu} = \gamma_5\gamma_\mu + \Gamma_{5\mu}\mathcal{G}_0\mathcal{K}$. The full dot is suppose to mark that the vertex is fully dressed and the empty dot for the bare vertex.

take the derivative with respect to x of this function we end up with

$$\begin{aligned}\partial^x_\mu G^\mu_{\gamma_\mu\gamma_5}(x_1, x_2, x) = &\langle 0|T(\partial_\mu j^\mu_{5\mu}(x))\psi(x_1)\bar{\psi}(x_2)|0\rangle \\ &+ \delta(x^0 - x^0_1)\langle 0|\mathrm{T}\left[j^0_{5\mu}(x), \psi(x_1)\right]\bar{\psi}(x_2)|0\rangle \\ &+ \delta(x^0 - x^0_2)\langle 0|\mathrm{T}\psi(x_1)\left[j^0_{5\mu}(x), \bar{\psi}(x_2)\right]|0\rangle.\end{aligned} \tag{2.34}$$

The commutator relation are known from the current algebra and if we insert these in eq. (2.34), we obtain the axial vector Ward-Takahashi identities(**AVWTI**):

$$P_\mu G_{5\mu}(q, k) = S^{-1}(q) i\gamma_5 + i\gamma_5 S^{-1}(k) - 2im(\mu^2) G_5(q, k), \tag{2.35}$$

where $G_5(q, k)$ is the **pseudo-scalar** 3 point-function, $m(\mu^2)$ the renormalized current quark mass and $S(k)$ the fully dressed quark propagator. The pseudo-scalar function G_5 is given by eq. (2.33), but including the pseudo-scalar current $j^5_a(x) = \bar{\psi}(x)\gamma^5 t_a \psi(x)$. The function obeys a homogeneous BSE. The renormalized mass is related to the bare quark mass m_0 appearing in the Lagrangian through the renormalization constant: $m(\mu^2) = Z_2(\mu^2) Z_4^{-1}(\mu^2) m_0 = Z_m^{-1} m_0$. The renormalized mass depends on the renormalization point μ. In this thesis we will work at a renormalization point of $\mu = 19GeV$ in the MOM scheme. Details on the renormalization scheme will be explained in the next section as part of the quark DSE. The AVWTI tell us how the quark propagator (two-point function) is related to the axial-vector three-point function. In the chiral limit, $m(\mu^2)^2 = 0$, the AVWTI reduces to

$$P_\mu \Gamma^j_{5\mu}(q, P) = S^{-1}(q_+)\gamma_5 \frac{\tau^j}{2} + \gamma_5 \frac{\tau^j}{2} S^{-1}(q_-). \tag{2.36}$$

where $S^{-1}(q_+)$ is the inverse quark propagator evaluated at momenta $q_\pm = q \pm \sigma P$, with partition parameter $\sigma \in \{0, 1\}$.

Every Greens function can generally be decomposed in terms of basis function τ_i times scalar dressing functions f_i:

$$G^{(n)} = \sum_{i=1}^{n} f_i \tau_i \tag{2.37}$$

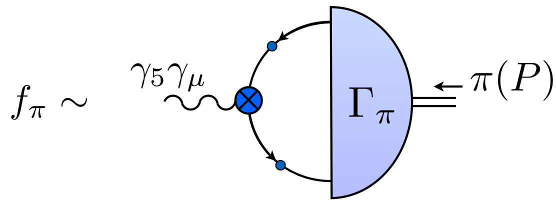


Figure 2.4: Diagrammatic definition of the pion decay constant f_π. The axial-vector vertex is coupled to the BSA for the pion.

The basis function reflects the quantum numbers of the object and are combinations of gamma matrices. The number of basis functions depends on the complexity of the Greens function. For the (2-point) quark propagator, for example, the basis consists of two basis functions. The particular basis for the quark and for the pion are given in section (3.2). Using the tensor decompositions and the AVWTI eq. (2.36) one obtains the chiral limit relation between the different dressing functions. A detailed derivation can be found in [5].

If we combine the axial-vector vertex three-point function with a pion bound state, we obtain an object proportional to the pion decay constant f_π, Fig.(2.4).

$$iP_\mu f_\pi = \langle \Omega | \psi \gamma_5 \gamma_\mu \bar{\psi} | \pi^0(P) \rangle \tag{2.38}$$

Where $f_\pi = 0.092$GeV and P is the on-shell momentum of the pion. We will use this relation as a cross-check on the result of the BSE equation in section (3.2). With the normalized pion BSA Γ_π we should be able to obtain the physical value for f_π.

Away from the chiral limit $m \neq 0$, the AVWTI contains an additional term $\Gamma_5^j(q,P) \sim G_5$, the pseudo-scalar 3-point (vertex) function, as can be seen in the full AXWTI eq. (2.35). This Greens function can also be decomposed in terms of its basis function and if we proceed similar to the chiral limit case, as described in [5] we obtain the Gell-Mann-Oakes-Renner (**GMOR**) relation:

$$f_\pi M_\pi^2 = 2m\ r_P, \tag{2.39}$$

The residue r_P are part of the decomposition for the pseudo-scalar vertex Γ_5. At this point the equation only tells us that in the chiral limit $(m(\mu^2) = 0)$ either $f_\pi = 0$ or $r_P = 0$. If we use the inhomogenous BSE for the pseudo-scalar vertex close to the pion pole and evaluate it in the chiral limit we find that the residue equals the quark chiral condensate $r_p = -\langle \psi\bar{\psi} \rangle / f_\pi$. Thus GMOR in the chiral limit

$$f_\pi^2 M_\pi^2 = -2m \langle \psi\bar{\psi} \rangle. \tag{2.40}$$

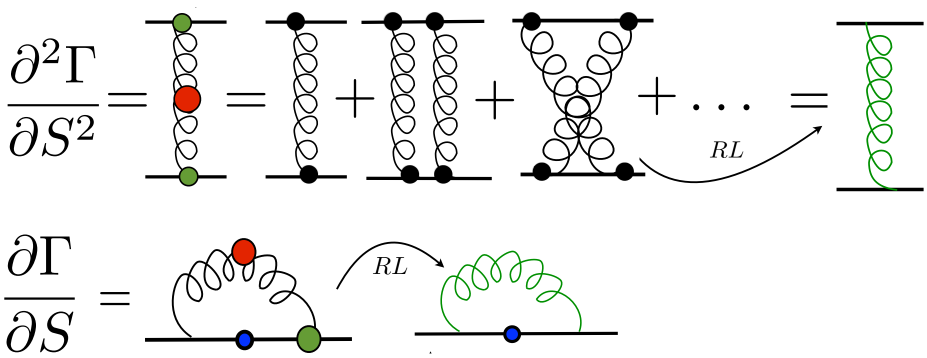


Figure 2.5: The first few terms of the skeleton expansion of the quark anti-quark interaction kernel $\mathcal{K}^{(4)}$. The arrow indicates the approximation of the kernel using the rainbow-ladder truncation. The quark-gluon vertex and the gluon propagator are replaced by an modeled effective interaction, check the detail on the rainbow-ladder truncation in section (3.1).

GMOR tells us that for a quark mass going to zero, the pion mass will go to zero as well. The decay constant and the condensate are good order parameters for chiral symmetry breaking, since they both vanish if chiral symmetry is not explicitly or dynamically broken.

2.7 Rainbow-ladder truncation

In the chapter where the Dyson-Schwinger equations were introduced, we already mentioned that the DSE is a set of infinite coupled integral equation that can only be solved by introducing a truncation on the system. This truncation needs to obey the symmetry relations of the theory in order to ensure the right physical result.

The effective action eq. (2.6) is a strong tool for systematically deriving interaction kernels. This ensures that all relevant symmetries of the system are maintained even when approximations are made. A detailed outline about how it works can be found in [6]. In this reference the authors describe that the 2-point interaction kernel $\mathcal{K}$ as well as the quark self-energy Σ can be obtained by taking functional derivatives of the effective action with respect to the propagator. Thus the self-energy and the kernel are connected and a systematic truncation has to be consistent in both.

The rainbow-ladder (RL) truncation approximates the interaction kernel as the exchange of a single gluon between quark and antiquark, as shown diagrammatically in Fig.(2.5). The quark-gluon vertex and the gluon propagator are modeled by an effective interaction developed by Maris and Tandy [7] and will be introduced in section(3.1). The kernel is the second derivative with respect to a fully dressed propagator S of the

2-PI effective action. This truncation is the simplest set-up to satisfy the **AVWTI**. A truncation not obeying the AVWTIs would not give the right physical prediction, such as ensuring that the pion in massless in the chiral limit (GMOR).

Attempts have been and are made to go beyond RL [8–11]. These papers try to obtain a model independent truncation for the kernel by including more interaction terms. The guidelines for such constructions are again given by the symmetries and symmetry relations of QCD, such as the Axial Ward-Takahashi identities. Effects seen in these calculations seem to be further off experimental values than RL calculations, but there are recent exceptions [12]. The discussion leads to interesting insight around the RL truncation and is of current research interest.

Nonetheless the result for different light meson spectra or the electroweak form factors of π and K using DSE in RL truncation are shown to be within 10-20 % of the experimental result. Therefore effects beyond RL are expected to be small and we trust the truncation for the investigation of pion decays in this thesis. While results for the ground-state pseudoscalar and vector mesons and ground state bayrons are accurate, the approximation shows clear deficiencies for excited states, states with finite width or mesons with axial-vector or scalar quantum numbers. The calculation can be found in [13–16]

3 The $\pi \to \gamma\gamma$ form factor and it's ingredients

During the days of Yukawa and colleagues, QCD was just in its beginning. It was already believed that we should consider four fundamental forces to describe the interactions in our universe, which are still those that we know of today: gravity, the electromagnetic force, the strong force and the weak force. It was experimentally demonstrated that the electromagnetic force is mediated by a particle now well known as the photon. In 1934 Yukawa predicted such a mediator particle for the strong force, which was later discovered and named the pion in 1947. As experimental techniques improved it became clear that the pion is not actually a fundamental particle after all. The mediator particle in case of the strong force was found to be a new particle called the gluon. Nevertheless some processes are still described fairly well using the pion as a mediator, as it is done in Chiral perturbation theory (χPT). In this context the $\pi\gamma\gamma$ vertex is an interesting quantity and has been calculated in many different approaches. For example using (χPT) [17], dispersive approaches [18] or with lattice gauge theory [19].

Luckily, for on-shell photons and massless pions, the form factor can be exactly calculated using the $SU(3)$ flavor axial anomaly. This is great because we can take the limit of the result obtained by the non-perturbative calculation and compare it with the exact perturbative result. To describe virtual photons and massive pions we can use a perturbative expansion in powers of $\mathcal{O}(m_\pi^2, p_1^2, p_2^2)$. The result from the anomaly, given by the 1-loop fermionic triangle diagram seen in Fig.(3.7), is the leading order term in this expansion. For physical pions and low virtuality photons the higher order corrections are suppressed. We can calculate higher order diagrams to estimate these corrections but all these diagrams vanish in the chiral and on-shell limit ($m_\pi, p_1^2, p_2^2 \to 0$). Thus in this limit the the triangle diagram gives the exact result. The two Noether currents associated with chiral symmetry are the vector current $J^a_\mu = \bar{\psi}\gamma^\mu\psi$ and the axial current $J^a_{\mu 5} = \bar{\psi}\gamma^5\gamma^\mu\psi$. These were introduced above in connection with the QCD flavor symmetries $SU(3)_V$ and $SU(3)_A$, as can be seen in the current table in chapter (2.6). The field equations of motion imply that the vector symmetry is conserved ($\partial_\mu J^\mu = 0$) and the axial symmetry is broken, as can be seen in the tabel of section (2.6.1) which I would like to quote here again for convenience:

$$\partial_\mu J^{\mu 5} = 2iM\bar{\psi}\gamma^5\psi - Tr[Q^2\lambda^a/2]e^2/(16\pi^2)\ \epsilon^{\mu\nu\alpha\beta}F_{\mu\nu}F_{\alpha\beta}. \tag{3.1}$$

In the chiral limit the first term in the divergence of the axial current vanishes, while the second term remains. This term appears because the axial symmetry is anomalous. More specifically, this effect is referred to as the Alder-Bell-Jackiw anomaly (ABJ-anomaly). This second terms comes mainly from the triangle diagram and the diagram is therefore often referred to as the anomaly diagram.

A symmetry is said to be anomalous if the symmetry of the classical action is not the symmetry of the quantum theory. In the chiral limit and for on-shell photons the ABJ anomaly predicts the form factor $\mathcal{F}_{\pi^0\gamma\gamma}$ exactly to be:

$$\mathcal{F}_{\pi^0\gamma\gamma}(m_\pi^2 = 0, p_1^2 = 0, p_2^2 = 0) = \frac{1}{4\pi^2 f_0} \tag{3.2}$$

where $f_0 = 0.092\text{GeV}$ is the pion decay constant f_π in the chiral limit. The total decay rate of the $\pi^0 \to \gamma\gamma$ decay at leading order in QED is given by:

$$\Gamma(\pi^0 \to \gamma\gamma) = \frac{\alpha_e^2 m_\pi^3}{16\pi^2 f_\pi}\, \mathcal{F}^2_{\pi^0\gamma\gamma}(m_\pi^2, 0, 0) = \frac{\alpha_e^2 m_\pi^3}{64\pi^3 f_\pi^2} \tag{3.3}$$

where $\alpha_e = e^2/4\pi \approx 1/137$ is the fine-structure constant and m_π is the neutral pion mass. Expressed in numbers, using the $\pi\gamma\gamma$ form factor in the chiral limit, the decay rate is given by $\Gamma(\pi^0 \to \gamma\gamma) = 7.77\text{eV}$. This is consistent with the experimental value of $7.73 \pm 0.16\text{eV}$.

The perturbative calculation of this form factor is also historically relevant since its result was one of the first confirmations that the numbers of colors in QCD had to be three. The decay rate was originally calculated starting with a QED Lagrangian with a Yukawa coupling between a fermion ψ, which was identified with the proton, and a pseudo scalar π. The Yukawa coupling was constrained by matching it to the Chiral Lagrangian. The strict calculation using the chiral Lagrangian is somewhat ambiguous, as can be seen by comparing the books Refs. [20] and [21]. The calculation of the one loop triangle diagram in this theory gives the exact answer. Since the final result is independent of the mass of the particle going around the loop, we can take it to be either quarks or protons. If we consider quarks, the factor of α_e^2 in eq. (3.3) must be multiplied by additional factors of electric charge squared and azimuthal component of isospin. Which leave us with an additional factor of $N_c/3$ in eq. (3.3). Since the result obtained without that factor in eq. (3.3) was so close to experimental values it was concluded that $N_c/3 = 1$.

The neutral pion form factor is not only an interesting quantity by itself, but is also interesting as a building block in describing other processes. An example are other, less frequent decays of the neutral pion, as we will discover in the following section. The form factor is also part of the light-by-light contribution to the g-2 calculation of the muon. Here both the theory and the experimental values are known to parts per million precision and there is a three sigma tension between them. One of the leading uncertainties on the theory number is the hadronic light-by-light contribution, in which three photons attach to the muon propagator and the fourth is external. This is expected to be dominated by a pion pole and for this reason the $\pi\gamma\gamma$ form factor is of great interest. The review [22] and the more recent [23] review summarize the history and progress in experimental and theoretical studies for the $(\pi^0\gamma\gamma)$ decay. Both include a discussion about the set-up of the recent PrimEx experiment at JLab and [23] shows a comparison of experimental data over the last years.

This chapter is organized as follows. First I will introduce all the ingredients of the neutral pion decay, explain the general structure and how they are calculated in the Dyson-Schwinger framework. Afterward everything will be put together for the form factor calculation. We conclude with the mathematical expression for the form factor.

3.1 Quarks

A central quantity of all calculations in this thesis is the fully-dressed quark propagator. It is the 2-point Greens function of quark fields and is calculated iteratively solving the Dyson-Schwinger equation (DSE) eq. (2.17) :

$$S(p) = G^{(2)}(p) = \int d^4x e^{-ipx} \langle \Omega | \mathrm{T}\ \bar{\psi}(x)\ \psi(0) | \Omega \rangle. \tag{3.4}$$

In the following section I will present the structure of the propagator in all relevant spaces (color, flavor, dirac) and show representative results. Starting with the quark evaluated in the real plane, followed by the calculation in the complex plane.

The color and flavor structure of the quark are rather simple. A quark propagating in spacetime with only self interaction will not change its color. Thus it will be proportional to the identity matrix in color space. The same is true in flavor space. With specifically written out flavor and color structure the quark propagator is:

$$S_{\alpha\beta abAB}(p) = S_{\alpha\beta}(p) \otimes \delta^{ab} \otimes \delta^{AB}, \tag{3.5}$$

with the color indices a, b, flavor indices A, B and the Dirac indices α, β. In general we can decompose the Dirac structure in terms of a basis τ_i times scalar dressing functions $f_i : \sum_i f_i \tau_i$. The construction of a basis in this space is done simultaneously using the gamma matrices with Lorenz index γ^μ and depends on the quantum number of the state. The quark propagator is a Dirac spinor. The two basis element are $\tau_i = \{\mathbb{I}_D, p^\mu \gamma_\mu\}$ and the scalar dressing functions, the scalar $\sigma_s(p)$ and the vector dressing function $\sigma_v(p)$:

$$S_{\alpha\beta}(p) = -i\sigma_v(p)\not{p} + \sigma_s(p)\mathbb{I}_D. \tag{3.6}$$

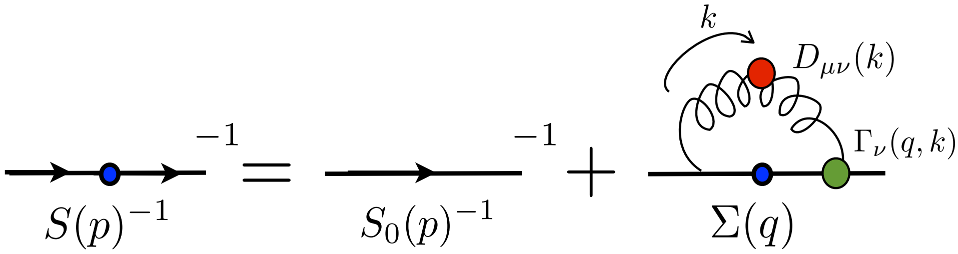


Figure 3.1: The Dyson-Schwinger equation for the quark propagator. The dots express that the quantity if fully dressed. The inverse quark propagator is given through the inverse bare propagator S_0^{-1} plus the quark self-energy Σ.

The DSE is defined with the inverse propagator. Sometimes rather the inverse propagator is directly defined, using two different dressing functions A and B, which can be related to the scalar and vector dressing functions.

$$S(p)^{-1} = i\not{p}A(p^2) + B(p^2)\mathbb{I}_D. \tag{3.7}$$

The inverse free quark propagator is given by $S_0^{-1}(p) = i\not{p} + m\mathbb{I}_D$, with the quark mass m. The DSE is the quantum equation of motion and as introduced in section(2.3) is given in euclidean space by:

$$S^{-1}(p^2) = S_0^{-1}(p^2) + \int \frac{d^4q}{(2\pi)^4}\, D^{ab}_{\mu\nu}(p-q)\left(g\frac{\lambda^a}{2}\gamma_\mu\right) S(q^2)\left(g\Gamma^b_\nu(q,p)\right), \tag{3.8}$$

where $D_{\mu\nu}$ is the dressed-gluon propagator, $\Gamma^a_\nu(p,q)$ is the quark-gluon vertex, S_0 the bare propagator, g the coupling constant, q the loop momentum and $k = p - q$ the gluon momentum as indicated in Fig.(3.1). The color indices are denoted by a, b and λ^a are the Gell-Mann matrices in flavor space. The Gell-mann matrices are the generators of $SU(3)_C$. The second term on the rhs is the so called self-energy Σ of the quark, as can denoted in Fig.(3.1).
Using the tensor basis as projectors we can separate eq. (3.8) into two coupled integral equations one for each dressing function A, B. To do so we multiply eq. (3.8) with the projectors $P_1 = \mathbb{I}/4 \otimes \delta^{ab} \otimes \delta^{AB}$ for the B equation and $P_2 = (-\not{p}/p^2) \otimes \delta^{ab} \otimes \delta^{AB}$ for A and take the trace. The trace is taken over all spaces: color, flavor and Dirac.

Renormalization

Taking a closer look we notice that the loop integral we are approaching in the DSE is a divergent integral. In many field theoretic calculations the integrals are divergent and

a systematic strategy of how to deal with divergences has been developed: renormalization. A short intro about the general stories of renormalization and the scheme used for working with the DSEs can be found in appendix (7.3.5). In this section I want to show the renormalization in case of the quark DSE.
For the DSE we will regularize using a cutoff and renormalize in the momentum-subtraction-scheme (MOM). We introduce a cut-off by restricting the loop k^2 integral limits to an infrared (IR) cutoff ε and to an ultraviolet (UV) cutoff Λ. The dressing functions and the bare mass will thus depend on the UV cutoff. If the UV cutoff is changed these parameters need to change as well in order to keep physical observables unchanged. The bare mass m_0 is related to the renormalized mass m through the renormalization constant Z_m:

$$m_0(\Lambda^2) = Z_m(\mu^2, \Lambda^2)m(\mu^2). \tag{3.9}$$

We introduce renormalization constants for every vertex, mass and dressing function in the theory. The renormalized and the bare quark propagator $S(p^2)$ are related by:

$$S(p^2, \Lambda^2) = Z_2(\mu^2, \Lambda^2)S_R(p^2, \mu^2). \tag{3.10}$$

The relation for the dressing functions, $A(p^2)$ and $B(p^2)$ follow through the definition of the quark propagator in eq. (3.7):

$$A^{-1}(p^2, \Lambda^2) = Z_2(\mu^2, \Lambda^2)A_R^{-1}(p^2, \mu^2). \tag{3.11}$$

All renormalization constants are related to each other through the Slanov-Taylor identities (STI):

$$Z_{1F} = \frac{Z_2}{\tilde{Z}_3} \quad , \quad Z_g\tilde{Z}_3 Z_3^{1/2} = 1\,, \tag{3.12}$$

with the quark-gluon vertex renormalization constant Z_{1F}, the quark-wave function constant Z_2, the ghost propagator constant $\tilde{Z}_3$, the charged renormalization constant Z_g, the gluon constant Z_3 and the ghost-gluon vertex constant $\tilde{Z}_1$ set to 1, compared to the STI given in the appendix eq. (7.57). Renormalizing the DSE with all constants taken into account eq. (3.8) becomes:

$$\begin{aligned} S(p^2, \mu^2)^{-1} &= Z_2(i\not{p} + Z_m m) \qquad (3.13)\\ &+ Z_g^2 g^2 \int_k^\Lambda Z_3 D^{ab}_{\mu\nu}(p-k, \mu^2)\left(\frac{\lambda^a}{2}\gamma_\mu\right) Z_2 S(k^2, \mu^2)\left(Z_{1F}^{-1}\Gamma^b_\nu(k, p, \mu^2)\right). \end{aligned}$$

$\int_k$ denotes the four-dimensional loop integral. The $(1/Z_3)$ factor comes form the gluon propagator and Z_{1F}^2 for the two quark-gluon vertices. The coupling constant g scales with $\sim 1/Z_g$, due to the relation of bare and renormalized coupling

$$g(\mu^2) = Z_g^{-1}(\mu^2, \Lambda^2)g(\Lambda^2). \tag{3.14}$$

Using the STI eq. (3.12) the factor in front of the self-energy multiplies together as

$$Z_2\left(Z_g^2 Z_3 Z_2 \frac{\tilde{Z}_3}{Z_2}\right) = \frac{Z_2}{\tilde{Z}_3}\cdot\left[Z_g^2\tilde{Z}_3^2 Z_3\right] = \frac{Z_2}{\tilde{Z}_3} = Z_{1F}. \tag{3.15}$$

The term in square brackets is renormalization group (RG) invariant and equals one, as can be seen using the STI, right eq. (3.12). With all remaining constant eq .(3.13) becomes

$$\begin{aligned} S(p^2,\mu^2)^{-1} &= Z_2(i\not{p} + Z_m m)^{-1} \\ &+ \frac{Z_2}{\tilde{Z}_3} g^2 \int_k^\Lambda D_{\mu\nu}^{ab}(p-k,\mu^2)\left(\frac{\lambda^a}{2}\gamma_\mu\right) S(k^2,\mu^2)\left(\Gamma_\nu^b(k,p,\mu^2)\right). \end{aligned} \tag{3.16}$$

From the relation of the bare and renormalized dressing functions, eq. (3.11), we know that $A(p^2,\Lambda^2)$ and $B(p^2,\Lambda^2)$ need to scale with Z_2. Therefore we multiply an additional Z_2 to eq. (3.8). In the end we only have two renormalization factors, Z_2 and Z_m in the DSE since the model introduced in the next section will carry the rest. Another constant is defined to be $Z_4 = Z_2 Z_m$. For $m \neq 0$ and momentum values in the UV region, we demand conditions on the renormalization constants to renormalize:

$$A(p^2=\mu^2) \stackrel{!}{=} 1 \quad , \quad B(p^2=\mu^2) \stackrel{!}{=} m(\mu^2). \tag{3.17}$$

At the renormalization point μ we want the A dressing function to be equal to one and B to equal the renormalized mass $m(\mu^2)$, thereby the mass function M will also equal the renormalized mass. For the renormalization constant this leads to

$$Z_2 = 1 - A(p^2=\mu^2,\Lambda^2) \quad , \quad Z_4 = 1 - \frac{B(p^2=\mu^2,\Lambda^2)}{m(\mu^2)}. \tag{3.18}$$

We solve the equation for the dressing functions by iteration. As we iterate we also have to enforce the renormalization conditions eq. (3.18).

Rainbow-ladder

As already mentioned in section(2.7) a truncation is needed for the quark self-energy term to reduce the DSE to a solvable problem. The truncation used for the quark self-energy is directly related to the kernel in the Bethe-Salpeter equation in the next section. This is because the rainbow-ladder (RL) truncation on the self-energy is in a sense a truncation on the effective action Γ, and the form of the interaction kernel $\mathcal{K}$ and the self-energy Σ are derived directly from Γ. The RL truncation approximates the gluon propagator and the quark-quark-gluon vertex with a bare vertex $\sim \gamma^\nu$ and an effective coupling term $\mathcal{G}(k^2)$. $\mathcal{G}(k^2)$ is suppose to encode the interaction between quarks and gluons and in the UV regime reproduces the QCD running coupling. RL respects the global symmetries of QCD such as chiral symmetry, Poincaré covariance and renormalization group invariance. Furthermore it produces vector and axial vertices that satisfy the respective WTIs, discussed above. In the axial case this leads to massless pions in the chiral limit. In the vector case in combination with the impulse approximation, it ensure the electromagnetic current conservation.

The quark self-energy in RL is given precisely by:

$$\frac{1}{Z_2\tilde{Z}_3} g^2 D^{ab}_{\mu\nu}(p-k)\Gamma^b_\nu(k,p) \to \mathcal{G}((p-k)^2)\frac{t_{\mu\nu}(p-k)}{(p-k)^2}\frac{\lambda^a}{2}\gamma_\nu, \tag{3.19}$$

with the transverse projector $t_{\mu\nu}(p) = \delta^{\mu\nu} + (p^\mu p^\nu)/p^2$ and the renormalization constants from eq. (3.12). The effective interaction $\mathcal{G}(k^2)$ is be modeled by an ansatz from Maris and Tandy (**MT**) [13]

$$\mathcal{G}(k^2) = \frac{D}{\omega^6} 4\pi^2 k^4 e^{-k^2/\omega^2} + \mathcal{F}_{UV}(k) \tag{3.20}$$

with

$$\mathcal{F}_{UV}(k) = \frac{4\pi^2\gamma_m(1-e^{-k^2})}{1/2\ \ln[e^2 - 1 + (1 + k^2/\Lambda^2_{QCD})^2)]}. \tag{3.21}$$

The constants in the model are $\Lambda_{QCD} = 0.234GeV$ and $\gamma_m = 12/(33-2N_f)$ with the flavor factor $N_f = 4$. D and ω are parameters that change for each set-up, e.g. for a renormalized mass at renormalization point $\mu = 19GeV$ of $m = 0.0037\ GeV$ we take $D = 0.93\text{GeV}^2$ and $\omega = 0.4\text{GeV}$. They are fitted to reproduce f_π and the chiral condensate. These are the only model dependent parameters that we introduce. In eq. (3.21), $\mathcal{F}_{UV}$ should be interpreted as an additional log-tail which provides the expected ultraviolet behavior of the effective running coupling, as predicted by the one-loop renormalization group equation. The IR behavior is modeled and constrained by phenomenology.

The model by Maris and Tandy used for the quark-gluon vertex and gluon propagator includes exactly the renormalization constants in the square brackets in eq. (3.15) and with these the model is renormalization point independent. Besides all constants includes in the MT model we are left with an overall factor of Z_2^2 in front of the self-energy term, as we will see in the following equations.

Using the rainbow-ladder truncation for the quark-gluon vertex and the gluon propagator, the color trace gives a factor of 4/3 on the right hand side. As the last step we take the Dirac trace and end up with the renormalized equations for the DSE dressing functions:

$$
\begin{aligned}
B(p^2) &= Z_4 m + Z_2^2\,\frac{4}{3}\int \frac{d^4q}{(2\pi)^4}\,\frac{3\cdot B(q^2)}{q^2A(q^2)^2+B(q^2)^2}\,\frac{\mathcal{G}(k^2)}{k^2}, \\
A(p^2) &= Z_2 + Z_2^2\,\frac{4}{3}\int \frac{d^4q}{(2\pi)^4}\,\frac{A(q^2)}{q^2A(q^2)+B(q^2)}\left[(pq)+2\frac{(pk)(qk)}{k^2}\right]\frac{\mathcal{G}(k^2)}{k^2},
\end{aligned}
\tag{3.22}
$$

with m being the renormalized mass here, since we used eq. (3.9) and the renormalized dressing functions.
The equation can be computed numerically using an iterative method. The result for a typical quark propagator with mass $m(\mu = 19\text{GeV}) = 0.0037\text{GeV}$ can be seen in Fig.(3.2). The interesting interesting observation is that even in the chiral limit the mass function of the quark $B(q^2)$ is non-zero. Which come due to the fact that part of the quark's mass is dynamically generated by the interaction via the strong force. This effect mainly appears in the IR region of momentum p^2 and can only be observed in non-perturbative calculations. As mentioned in section(2.6.2) this dynamically mass generation is an indicator for dynamical chiral symmetry breaking $D\chi SB$.

Results

In all of the following calculations we used a Pauli-Villars cutoff Λ_{PV}. This cutoff gets multiplied to the effective interaction of the gluon in the DSE and in the BSE, such that:

$$
\frac{\mathcal{G}(k^2)}{k^2} \to \frac{\mathcal{G}(k^2)}{k^2}\left(\frac{\Lambda_{PV^2}}{k^2+\Lambda_{PV}^2}\right)
\tag{3.23}
$$

In this case the parameter of the Maris-Tandy Model eq. (3.20) are chosen to be $\omega = 0.4$ and $D = 1.0$. The PV cutoff fixed the issues that arise due to our hard UV-cutoff Λ. Introducing a hard cutoff to the theory breaks gauge-invariance. Additionally using PV solves this problem.

The calculation for the real quark propagator can be seen in Fig.(3.2). The quark mass function $M(p^2) = B(p^2,\mu^2)/A/(p^2,\mu^2)$ is additionally plotted, this is a renormalization point independent function. In the calculation for the form factor a quark mass of $m = 0.0037$ GeV is used which is also shown in the figure.

3.1.1 Quark in the complex plane

To evaluate the quark in the complex plane we choose to let the complex momentum run through the gluon propagator. This implies that the integrated loop momentum is still real. For the value of the dressing function at any complex momentum we therefore only need to plug in this external momentum p into eq. (3.22) and perform the loop integral q one more time. The advantages of this approach are the simple calculation,

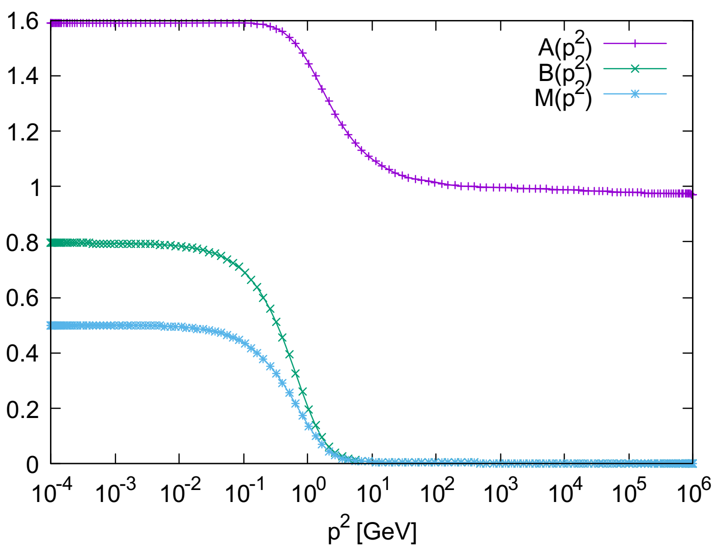


Figure 3.2: The quark dressing function and the quark mass function $M(p^2)$ calculated for a quark mass $m = 0.0037$GeV and with a Pauli-Villar cut-off: $D = 1.0$, $\omega = 0.4$. UV cut-off of $\Lambda^2 = 10^6$, IR $\varepsilon^2 = 10^{-4}$ and PV-cuttoff $\Lambda_{PV} = 2.0 \cdot 10^2$.

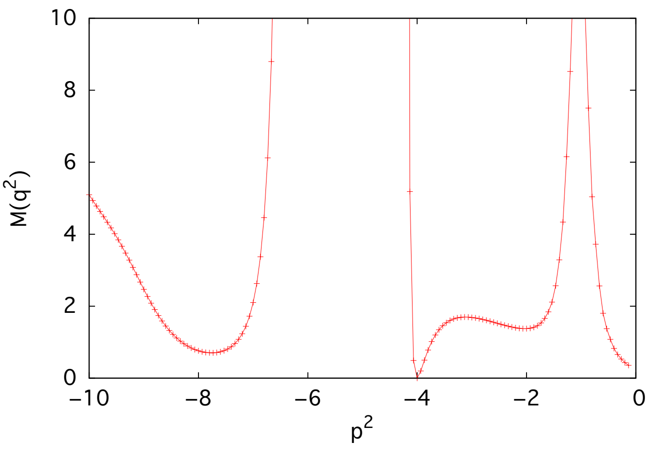


Figure 3.3: Plot of the complex quark mass function $M(q^2)$. The parameters for the calculation are: $w = 0.16GeV$, $D = 0.93GeV^{-2}$ and $m = 0.0037$ and without the PV cutoff.

but the downside is that we inherit numerical fluctuation due to the Maris-Tandy ansatz for the quark-gluon interaction and these errors are hard to estimate. Another option is to route the complex momentum though the quark propagator. In this case one would need to iteratively solve the DSE and interpolate in the complex plane. Uncertainties due to interpolation might appear, but at least these are numerical effects and are easier to estimate. The quark propagator received from the second method shows a smoother appearance and seems to be numerically more stable. Fortunately due to the rather small value of the pion mass the quark momenta needed in the calculation in this thesis are close to the real axis, thus we do not require momenta with large imaginary parts. In the following calculation the first method, routing the complex momentum through the gluon, was used.

The plot in Fig.(3.3) show the mass function $M(p^2)$ for time-like momenta. In our calculation we will need the quark propagator for space-like momenta and as mentioned close to the real axis. Thus it will only differ slightly from the real results in Fig.(3.2).

3.2 Meson

In the quark model mesons are quark anti-quark bound states, and this general idea is also a good guide for their nature within QCD. Such boundstates can be described through the 2-point bound state equation, called the Bethe-Salpeter equation (**BSE**), as introduced in section (2.4.1). The Bethe-Salpeter amplitude(BSA) Γ, is the object of interest, obtained by iteratively solving the BSE.

Color & Flavor

The BSA can be decomposed into the direct product of the structure in Dirac $\alpha\beta$, flavor AB and color space ab:

$$\Gamma^I_{\alpha\beta abAB}(p; P) = \Gamma^I_{\alpha\beta}(p, P) \otimes \delta^{ab} \otimes r^I_{AB}. \tag{3.24}$$

The matrix r_{AB} in flavor space represents each flavor structure of the different mesons, with $I = -1, 0, 1$ demoting the azimuthal component of isospin. This defines which member of the pion triplet we are studying. As discussed above, in this thesis we are only concerned with the neutral pion, π^0, with $I = 0$.

Throughout this thesis we work with two different quark flavors $\{u, d\}$, with the same masses. In this case the symmetry in the isospin space is not explicitly broken and up and down quark form a doublet under $SU(2)_F$. The theory behind these group representations is group theory and can be found in many good text books like [24]. Through group theory it is known that the irreducible representation of $2 \otimes 2 = 1_a \oplus 3_s$ is the product group of an anti-symmetric singlet and a symmetric triplet. The Pauli-matrices σ_i combined with the identity in 2×2 space turn out to be the appropriate basis. The flavor structure for in particular the pions is expressed through:

$$\begin{aligned} r^+ &= |u\bar{d}\rangle = \frac{1}{2}(\sigma_1 + i\sigma_2) && (3.25)\\ r^- &= |\bar{u}d\rangle = \frac{1}{2}(\sigma_1 - i\sigma_2) \\ r^0 &= \frac{1}{\sqrt{2}}\left(|d\bar{d}\rangle - |u\bar{u}\rangle\right) = \frac{1}{\sqrt{2}}\sigma_3 \\ r^s &= |u\bar{u}\rangle + |d\bar{d}\rangle = \frac{1}{\sqrt{2}}\mathbb{I}. && (3.26) \end{aligned}$$

I chose to take the matrices in color and flavor space unnormalized, therefore $r^0_{AB} = \sigma_3$. The meson, like all observable states in QCD, is a singlet in color. In other words it transforms as a singlet in the $SU(3)$ color space and is denoted by δ^{ab}.

Dirac & Lorenz

The particle we are interested in this thesis is the neutral pion, which is a pseudo-scalar. We use different basis representation for scalar, pseudo-scalar, vector, axial-vector mesons. The quantum number of a particle indicated the behavior under continuous Lorenz transformations. The pseudo-scalar π^0 has the quantum numbers $J^{PC} = 0^{-+}$. The 4-point bound-state equation for the meson depends on two independent momenta, the relative q momentum and the total momentum P. Similar to the quark propagator we try to form a basis obeying the pseudo-scalar behavior under transformation and the dependence on those two momenta. For the neutral pion an appropriate basis would be:

$$\begin{aligned} &\tau_1 = \gamma_5 && \tau_2 = -i\gamma_5 \not{P} && (3.27)\\ &\tau_3 = -i\gamma_5 \not{q}(P\cdot q) && \tau_4 = \gamma_5\left[\gamma^\mu, \gamma^\nu\right]. \end{aligned}$$

The $\pm i\gamma_5$ in front of all basis components ensures the minus sign under a parity transformation. A pseudo-scalar is odd under parity. Furthermore the neutral pion is a C-parity eigenstate. A C-parity transformation flips the sign of the relative momentum. All basis terms beside τ_3 are even under parity, to match this we need to multiply the third component with an additional scalar factor of $(P\cdot q)$. The normalized projectors P_i according to the basis can be constructed. Therefore we create a matrix with the entries $Tr[\tau_i \cdot \tau_j]$ with an additional row, which is a vector of the basis functions τ_i. This matrix is row reduced and the entries of the last row are the normalized projectors P_i, which follow as:

$$\begin{aligned} &P_1 = \frac{\gamma_5}{4} && P_2 = \frac{i\left(q^2\not{P} - P\cdot q\not{q}\right)\gamma_5}{4\left(P^2q^2 - P\cdot q^2\right)} && (3.28)\\ &P_3 = \frac{i\left(P\cdot q\not{P} - P^2\not{q}\right)\gamma_5}{4\left(P\cdot q^2 - P^2q^2\right)} && P_4 = \frac{\left[\gamma^\mu,\gamma^\nu\right]\gamma_5}{16\left(P\cdot q^2 . P^2q^2\right)}. \end{aligned}$$

The total expression for the neutral pion π^0 in tensor structures is

$$\Gamma^{I}_{\alpha\beta abAB}(p;P) = \sum_{i=1}^{4} f_i(q,P)\tau_i^{\alpha\beta}(q,P) \otimes \delta^{ab} \otimes r^{I}_{AB}. \tag{3.29}$$

The dressing functions f_i and the basis $\tau_i(q,P)$ depend on the four vectors for the total and relative momentum of the bound-state. If the pion is on-shell this reduced to two scalar variables, the angle between P and q and the magnitude of $|q|$.

As derived in section(2.4.1) eq. (2.28) the BSE for the BSA Γ reads

$$\Gamma^{I}_{\alpha\beta abAB}(p,P) = \int_k^{\Lambda} \mathcal{K}(P,q,p)^{\alpha'\beta'a'b'A'B'}_{\alpha\beta abAB} [S(q_+)\Gamma(P,q)S(q_-)]_{\alpha'\beta'a'b'A'B'} \tag{3.30}$$

with all color, flavor and Dirac indices explicitly written out. $\mathcal{K}$ is a quark anti-quark interaction kernel, defined by all diagrams that are 2PI in the s-channel. S the quark propagator introduced the section above. It is given diagrammatically in Fig.(3.4). The momenta are $q_+ = q+\sigma P$, $q_- = q-(1-\sigma)P$, with the distribution parameter $\sigma \in \{0,1\}$, and the total momentum $P_i = -M_\pi^2$ is fixed to the mass of the pion. The homogeneous BSE, eq. (2.28), as well as the DSE is self-consistent equation that we have to solve using iteration. If we numerically approximate the integral as a sum, then the homogeneous BSE becomes a standard eigenvalue problem

$$\Gamma_n = [\mathcal{K}\mathcal{G}_0]_{nm}\Gamma_m\,, \tag{3.31}$$

where n and m are collective indices for color, flavor, Dirac, structure and also the list of discrete momenta that we sum to evaluate the integral. This equation is only valid for $P^2 = M_\pi^2$. However, we can extend this equation for all P^2 by including a general eigenvalue, λ,

$$\lambda(P^2)\Gamma(P^2) = \big[\mathcal{K}(P^2)\mathcal{G}_0(P^2)\big]\Gamma(P^2)\,. \tag{3.32}$$

For a given input quark mass, one can solve eq. (3.32) for various values of P^2 and so determine $\lambda(P^2)$. For most values of P^2 this equation has no physical meaning. However, when $\lambda = 1$, then we know from eq. (3.31) that $P^2 = M_\pi^2$. Thus we can determine the pion mass by tuning P^2 until the eigenvalue becomes one.

Rainbowladder

In RL the BSE kernel $\mathcal{K}$ is replaced by an effective one-gluon exchange:

$$\mathcal{K}(q,p,P) \to Z_2^2 \frac{\mathcal{G}((p-q)^2)}{(p-q)^2} D^{free}_{\mu\nu}(q) \frac{[\lambda^i]_a^{a'}}{2} [\gamma_\mu]_\alpha^{\alpha'} \delta^{ij} \frac{[\lambda^j]_b^{b'}}{2} [\gamma_\nu]_\beta^{\beta'} \delta_A^{A'} \delta_B^{B'}\,, \tag{3.33}$$

where $\mathcal{G}(q^2)$ is a modeled effective coupling, the free gluon propagator is a transverse projector: $D^{free}_{\mu\nu}(k) = \big(\delta_{\mu\nu} - k_\mu k_\nu/k^2\big)$ and the vertex renormalization constant Z_2. λ^i are the Gell-Mann matrices as the gernerating matrices in flavor space with $i = 1\ldots 8$.

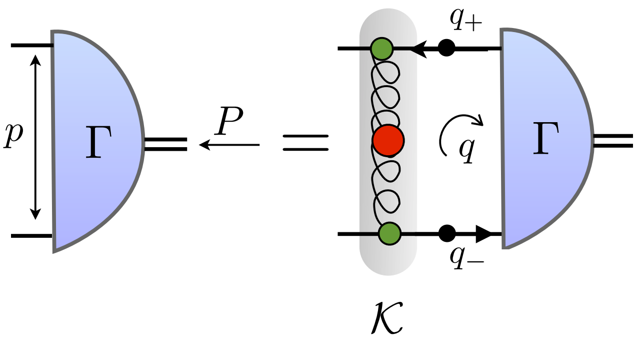


Figure 3.4: The homogenous Bethe-Saltpeter equation for the BSA Γ. With the total momentum off the bound-state P, the relative momentum between constituent quarks p and the loop momentum q. The Kernel will be approximated through the RL truncation.

In RL truncation the color trace of eq. (3.30) in detail

$$Tr_C\left[\delta^{\dagger}_{ab}\frac{[\lambda^i]^{a'}_a}{2}\delta^{ij}\frac{[\lambda^j]^{b'}_b}{2}\delta_{a'b'}\right]=\frac{4}{3}, \tag{3.34}$$

the trace in flavor space gives a factor of one. In total this concludes in the following equation for the neutral pion bound-state:

$$\Gamma(P,p)=-\frac{4}{3}\,Z_2^2\int\frac{d^4q}{(2\pi)^4}\frac{\mathcal{G}(q-p)^2}{(q-p)^2}\gamma_\mu S(q_+)\Gamma(q,P)_\pi S(q_-)\gamma_\nu \tag{3.35}$$

The amputated BSA Γ can be transformed into the Bethe-Salpeter wave-function (BSW) χ using a rotational matrix Y_{ij}. The BSW is convenient to use in the calculation of the pion form factor, because the trace in Dirac space simplyfies slightly. The rotational matrix and the BSE expressed through χ can be found in the appendix (7.4).

Normalization

The bound state amplitude Γ_π, obtained by solving the BSE, has an arbitrary over all normalization. We fix this by imposing the normalization given by the canonical normalization condition, first given in section eq. (2.30):

$$Tr_{DFC}[\delta^{ij}]\frac{1}{N^2}=\frac{P^\mu}{2P^2}\frac{d}{dp_\mu}\int_q^\Lambda Tr_{DFC}[\bar{\Gamma}^i_\pi(q,-P_i)S(q_+)\Gamma^j_\pi(k,P_i)S(q_-)], \tag{3.36}$$

where $q_\pm$ are equal their definition in the explanation below eq. (3.30), $P_i=iM_\pi$ is fixed the mass of the pion and N the normalization factor that we want to determine.

The total momentum P which is part of $q_\pm$ is still arbitrary and needs to be taken into account for the derivative. The trace is taken over flavor, color and Dirac space. Using the BSA Γ_π as shown in eq. (3.29), we obtain a factor of 6 in front the integral when taking the color and flavor trace. The conjugated amplitude $\bar{\Gamma}$ equals Γ, besides the sign switch in the total momentum. We know how the Dirac structure of the basis function τ_i for the pion amplitude transform under complex conjugation and thus we know $\bar{\Gamma}$. The normalization factor N is calculated after solving the iterative equation for the BSA and multiplied to the amplitude. The normalized and unnormalized BSA are related through: $\Gamma^N = \Gamma\sqrt{N}$

As a crosscheck for the pion BSA and the normalization constant N it is convenient to calculate the pion decay constant $f_\pi = 0.092\text{GeV}$. Using the normalized amplitude calculate the pion decay constant from the following identity

$$\delta^{ij} f_\pi P_\mu = Z_2 \int_k^\Lambda Tr_{DFC}[\frac{\tau^i}{2}\gamma_5\gamma_\mu S(k_+)\bar{\Gamma}^j_\pi(k,-P)S(k_-)]. \tag{3.37}$$

We introduced this relation via the axial-vector vertex in eq. (2.38) and Fig.(2.4) in the former section.

Results

Fig.(3.5) shows the dressing function of the pion for a quark mass $m = 0.0037\text{GeV}$ and the pion mass resulting from solving the BSE is $\mathbf{M_\pi = 0.1374GeV}$. A PV-cutoff was used in the calculation eq. (3.23) and the UV cut-off $\Lambda^2 = 10^6\text{GeV}^2$ and IR cut-off $\varepsilon^2 = 10^{-4}\text{GeV}^2$. From the chiral limit relations we know that the first dressing function of the pion BSA equals the second dressing function of the quark propagator divided by the pion decay constant $f_1(p,P) = B(p)/f_\pi$. For small quark masses we notice that the functions are still about the same, as plotted in Fig.(3.5). The normalization constant was calculated with $N = 8.73$ and with the normalized pion amplitude we obtain a pion decay constant of $f_\pi = 0.092\text{GeV}$.

An earlier result using the Dyson-Schwinger formalism can be found in [14]. In that work the authors give a detailed discussion about the renormalization concerning the DSE and solve the BSE for the pion and the kaon.

3.3 Quark-photon vertex

The quark-photon vertex satisfies the renormalized inhomogeneous BSE

$$\tilde{\Gamma}^\mu(P,p) = Z_2\hat{Q}\gamma_\mu + \int_q^\Lambda \mathcal{K}(q,p,P)S(q_+)\Gamma_\mu(q,P)S(q_-). \tag{3.38}$$

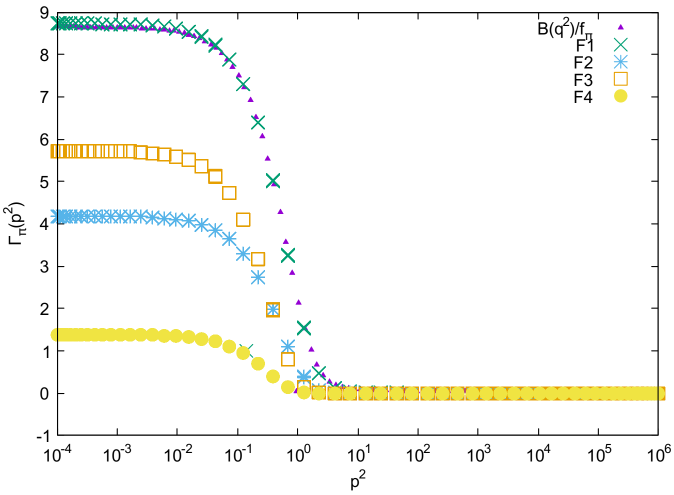


Figure 3.5: Dressing functions f_i of the pion BSA Γ_π with quark mass at m_q = 0.0037GeV. In the chiral limit it is known that the quark dressing function and the first pion dressing function are related through $B(q^2)/f_\pi^2 = f_1$. The plot shows $B(q^2)/f_\pi$ and we see that even away from the chiral limit for small quark masses the functions are about the same.

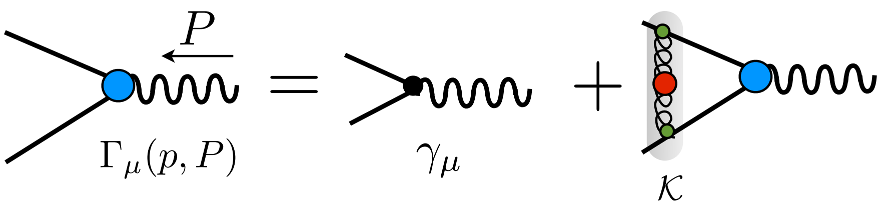


Figure 3.6: The inhomogeneous Bethe-Saltpeter equation for the quark-photon vertex.

The kernel $\mathcal{K}$ is the same amputated $q\bar{q}$ scattering kernel as in the homogeneous BSE for the pion amplitude and is 2PI in the s-channel. The momenta are determined according to the momentum distribution for the pion amplitude and $\hat{Q}$ is the quark charge operator.

Color & flavor

The quark-photon vertex is a direct product in color, flavor and Dirac space and is given by

$$\tilde{\Gamma}^{\mu}_{\alpha\beta abAB}(p,P) = \Gamma^{\mu}_{\alpha\beta} \otimes \delta^{ab} \otimes \delta^{AB}\tilde{Q}, \tag{3.39}$$

with $\tilde{Q} = diag(2/3, -1/3)$ the quark charge matrix in $SU(2)$.

Dirac & Lorenz

There are twelve invariant basis elements needed to represent all Dirac structures of the quark-photon vertex. Four of them are longitudinal and can be expressed in terms of the quark propagator dressing function via the vector WTI. The other eight transverse amplitudes are not constrained by the WTI, except at $P = 0$, and can be chosen as projectors of the form

$$\begin{aligned} \tilde{\Gamma}^{\mu}_{T}(p,P) = \quad & \gamma_\mu \quad V_1 + \not{p} p_\mu V_2 + p_\mu \not{P} V_3 \\ + \quad & \gamma_5 \epsilon_{\mu\nu\alpha\beta}\gamma_\alpha p_\nu P_\beta V_4 + p_\mu V_5 \\ + \quad & \sigma_{\mu\nu} p_\nu V_6 + \sigma_{\mu\nu} P_\nu V_7 \\ + \quad & p_\nu \sigma_{\alpha\beta} p_\alpha P_\beta V_8. \end{aligned} \tag{3.40}$$

The quark-photon vertex can be separated into a 'gauge part' and a purely transverse part in eq. (3.40)

$$\tilde{\Gamma}^{\mu}(p,P) = \tilde{\Gamma}^{BC}_{\mu}(p,P) + \tilde{\Gamma}^{\mu}_{T}. \tag{3.41}$$

The gauge part is the so called Ball-Chiu Ansatz and it is expressed in terms of the quark dressing functions $A(k)$ and $B(k)$

$$\tilde{\Gamma}^{BC}_{\mu}(p,P) = \hat{Q}\left[\frac{1}{2}\gamma_\mu(A(p_+) + A(p_-)) + 2\not{p} p_\mu \frac{A(p_+) - A(p_-)}{p_+^2 - p_-^2} - 2ip_\mu \frac{B(p_+) - B(p_-)}{p_+^2 - p_-^2}\right]. \tag{3.42}$$

The BC part of the vertex already satisfies the WTIs, transforms correctly under CPT and has the right perturbative limit of γ_μ in the ultraviolet region. It completely describes the longitudinal part of the vertex, but this part is missing the expected vector meson pole behavior. The transversal part of the vertex behaves close to the ρ pole like

$$\tilde{\Gamma}_{\mu}(p,P) \sim \frac{\Gamma^{\rho}_{\mu}(p,P) m_\rho^2/g_\rho}{P^2 + m_\rho^2}, \tag{3.43}$$

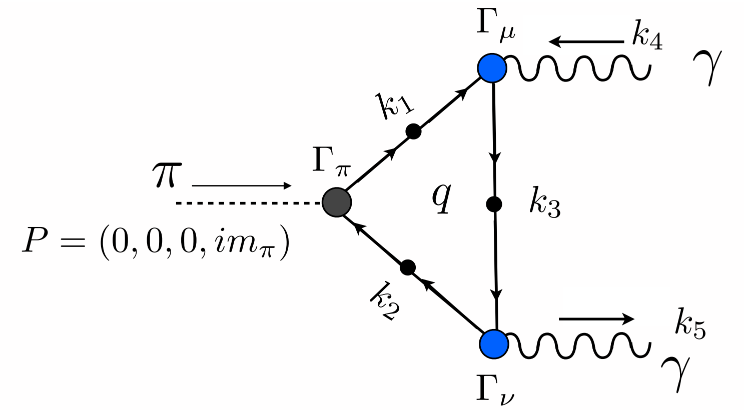


Figure 3.7: Triangle diagram for the ($\pi^0 \to \gamma\gamma$) decay. The momentum of the incoming pion is chosen to be on-shell and in rest. q is the loop momentum. The relative momentum between the photon is defined by the photon momenta k_4 and k_5 through $p = k_4 + k_5$ and the total momentum $P = k_5 - k_4$.

where g_ρ is the coupling constant and the mass m_ρ.

Solving the vertex with all twelve tensor structures is a complex calculation and was thoroughly performed using the Dyson-Schwinger methods in [25]. In this paper the author calculated the full quark-photon vertex in RL and parametrized the eight transverse dressing functions. This parametrization is used in the calculation for the form factor here and the detailed structure is explained in the next paragraph.

3.4 The form factor

Now that we assembled all ingredients we can compute the $\pi^0 \to \gamma\gamma$ form factor. The decay amplitude of the form factor can be decomposed into tensor structures times scalar functions, as applied in the calculation of the pion amplitude. The only two-Lorentz-index tensor structure with the right parity transformation is the Levi-Civita Tensor contracted with the four-vectors of the two photons:

$$F_{\mu\nu}^{\pi^0\gamma\gamma}(P, p) = \varepsilon_{\mu\nu\alpha\beta} k_4^\alpha k_5^\beta \mathcal{F}_{\pi^0\gamma\gamma}(p, P). \tag{3.44}$$

Here the scalar amplitude $\mathcal{F}_{\pi^0\gamma\gamma}$ depends on the total momentum P and the relative momentum between the photons p. The particular momentum distribution can be found in the appendix (7.1), k_4 and k_5 are the photon momenta as can be seen in Fig.(3.7).

In the case of off-shell photons and away form the chiral limit we cannot expand the form factor in terms of a small variable in a series of diagrams, i.e. we cannot use perturbation theory. The QCD coupling constant is known to be large in this region

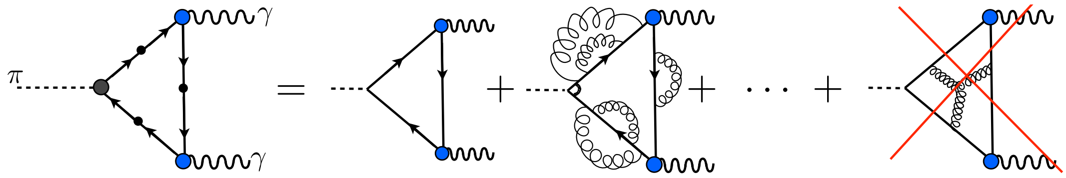


Figure 3.8: The figure shows the scheme of diagrams that are taken in account for the decay using the rainbow-ladder truncation. Higher order correction to the quark propagator and the pion amplitude are included in the RL scheme. Higher vertices correction however, like a three gluon vertex, presented by the diagram on the far right, are not included.

and also the pion mass $M_\pi \neq 0$ so results calculated in the chiral limit are not directly applicable. Therefore we need a different method to calculate the full form factor. A useful tool for describing electromagnetic processes such as $\pi^0 \to \gamma\gamma$ is the impulse approximation. This approximation takes only the leading order diagram of the process into account, but takes all quantities in this tree level diagram to be fully dressed and thus includes higher order interactions.

This is consistent with the rainbow-ladder truncation. The RL truncation implies the impulse approximation. Working in RL the one-loop triangle diagram is the only diagram considered for the decay. The RL truncation does not include diagrams with higher order vertices. This means that all self-interactions of the quark are taken into account when using fully dressed quark in RL, but diagrams which couple simultaneously to different quark lines are not included, as can be seen in Fig.(3.8). For example, the three-gluon vertex with one gluon attaching to each side of the triangle is not included.

We can introduce a rescaled form factor, such that this form factor equals one in the on-shell chiral limit $F_{\pi^0\gamma\gamma}(P^2 = 0, p^2 = 0, P \cdot p = 0) = 1$:

$$\mathcal{F}_{\pi^0\gamma\gamma} = \frac{e^2}{4\pi^2 f_\pi} F_{\pi^0\gamma\gamma}(P, p). \tag{3.45}$$

In the impulse approximation, the pseudoscalar to two-photon vertex for the quark flavor content of the pion, $\{a\bar{b} = u\bar{u}, d\bar{d}\}$, is given by

$$F^{ab}_{\mu\nu}(P, p) = 2N_C \int_q \Gamma^{ab}_\pi(P, p) S^{bc}(k_1) ie\tilde{\Gamma}^{cd}_\nu(P, p) S^{de}(k_3) ie\tilde{\Gamma}^{ef}_\mu(P, p) S^{fa}(k_2)], \tag{3.46}$$

where the roman letters are flavor and Dirac indices and the trace in color space is already taken and results in a factor of $N_C = 3$ up front. The factor 2 is a symmetry factor: There are two diagrams contributing to the decay. The second diagram has the same topology as Fig.(3.7) but the photon legs are exchanged. Γ_π is the pion Bethe-Salpeter amplitude, already introduced above, and $\tilde{\Gamma}_\mu$ the quark-photon vertex. The pion amplitude is normalized as described above. Taking the flavor trace leads to a factor of

$$\left[r_0^{ab}\tilde{Q}^{bc}\tilde{Q}^{ca}\right] = Tr_F\left[\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}2/3 & 0\\ 0 & -1/3\end{pmatrix}\begin{pmatrix}2/3 & 0\\ 0 & -1/3\end{pmatrix}\right] = \frac{1}{3}, \tag{3.47}$$

which cancels the color factor and leaves an overall factor of 2 in front of the integral. Since the form factor is calculated for the pion on its mass-shell, the momenta dependence is simple.($F_{\pi\gamma\gamma}(|p|, \hat{Pp})$ does only depend on two variables: the magnitude of the relative momentum p between the out-going photons and the angle between this relative momentum and the total momentum P, with the angle $\hat{Pp} = p \cdot P/\sqrt{p^2P^2}$.

4 Results & Comparison

The recent PrimEx experiment at JLab has measured the neutral pion decay width with an accuracy of 2.8%. At this level of accuracy it is interesting to know the correction of the form factor due to finite pion mass. Hence from a theoretical point of view it is interesting to know the ($\pi^0 \to \gamma\gamma$) form factor away from the chiral limit. The value of the form factor for off-shell photons is also interesting for the reasons discussed above. In particular there are less frequent decay processes of the pion that can only be determined with non-perturbative knowledge of this from factor. Another motivation is the calculation of the hadronic light-by-light scattering, which is the second largest theoretical error in the determination of the muon $g-2$. The off-shell form factor can only be calculated through the use of **non-perturbative** methods. The advantage compared to the perturbative calcuation, is that the full effects of the non zero pion mass and off shell momenta are included. Two of the most promising methods are numerical lattice QCD and the **Dyson-Schwinger** (**DSE**) formalism. Attempts have also been made to make a prediction for the off-shell form factor in terms of the vector meson dominance model (VMD) [26], using chiral perturbation theory (χPT) [17] or dispersive approaches [18].

$\pi^0 \to \gamma\gamma$

For the on-shell pion form factor the photon momenta $k_4^2 = k_5^2 = 0$ and we obtain the following value for the normalized decay amplitude

$$\boxed{F_{\pi^0 \to \gamma\gamma} = 0.98941 \qquad (4.1)}$$

This concludes in a total decay rate of $\Gamma(\pi^0 \to \gamma\gamma) = 7.618\text{eV}$ while the experimental value of [$\Gamma = 7.8\text{eV}$ or $F_{\pi\gamma\gamma} = 1.012$] and the value obtained from the SU(3) axial anomaly $\Gamma = 7.7\text{eV}$. Which is a discrepancy of 2.3%.

$\pi^0 \to \gamma^*\gamma$

The ($\pi\gamma^* \to \gamma$) transition form factor, with one off-shell and one on-shell photon, was measured by several collaborations over the last 20 years [27–30]. The CELLO [28] collaboration at DESY first measured $\mathcal{F}_{\gamma^*\gamma\pi^0}$ at a space-like squared momentum transfer Q^2 up to 2.9 GeV^2. Later the CLEO collaboration [27] result followed, measuring in regions of momentum transfer between 1.5 and 9 GeV^2 and more recent results from

BaBar [29] or BELLE [30] go up to a region of $Q^2 \sim 40\text{GeV}^2$. The decay is observed as part of e^+e^- scattering. To leading order in QED e^+e^--scattering can be described as the interaction between two photons emitted by the scattered electron. These photons can produce a pair of strongly interacting quarks, which are observed in the form of hadrons. The CLEO collaboration measured the $e^+e^- \to e^+e^-\mathcal{R}$ cross section, where $\mathcal{R}$ is either a π^0, η or η'. The measurement was done in a "single-tagged" experimental mode, which means that only one of the scattered electrons is detected, while the other is scattered at a small angle. In this case the photon emitted by the "tagged" electron is an off-shell photon (γ^*), whereas the momentum transfer to the photon emitted by the other electron is nearly zero, and the photon is therefore on-shell. To relate transition form factors from the differential cross section the process is divided into a perturbative part $e^+e^- \to e^+e^-\gamma^*\gamma$ and a non-perturbative part $\gamma^*\gamma \to \pi^0$. The first part is completely calculable in QED and the second part is defined as the transition form factor $\mathcal{F}_{\gamma^*\pi\gamma}$. For the analysis of the data a Monte-Carlo (MC) simulation program was used.

The ($\pi^0 \to \gamma^*\gamma$) was not directly calculated in course of this thesis, since the important input for the next calculation of other pion decay is the off-shell form factor. But the results of CLEO and CELLO have been compared to DSE calculations in [31] and found to agree. The dependence on the transferred momentum squared Q^2 of the form factor was studied within all measurements and found to be slightly different, especially in the higher Q^2 region. The BaBar collaboration observed a $\sim 1/Q^{3/2}$ dependence while BELLE's results seem to agree with the leading order result of perturbative QCD, $\sim 1/Q^2$. This calls for more precise theory predictions of the form factor in higher momentum ranges.

$\pi^0 \to \gamma^*\gamma^*$

The off-shell form factor $\mathcal{F}_{\pi\gamma^*\gamma^*}$ corresponds to a pion on-shell decaying into two virtual photons. Even though this decay cannot be directly measured in experiment it is an interesting quantity form a theoretical point of view. The result of the off-shell form factor will be used as an input for other pion decay, in the following section(5). The result of the form factor plotted against the two virtual photon momenta can be seen in Fig.(4.1). The form factor is compared against the Vector-Meson-Dominance (VMD) model quoted in [26]. The VMD describes the form factor as

$$F_{\pi^0\gamma^*\gamma^*} = \frac{M_\nu^2}{k_4^2 + M_\nu^2}\frac{M_\nu^2}{k_5^2 + M_\nu^2}, \tag{4.2}$$

where $M_\nu = 0.769$ GeV, the mass of the ρ meson. In the symmetric case the VMD leads to dipole behavior, because both photons behave like $1/(p^2 + M_\nu^2)$ close to the ρ pole.

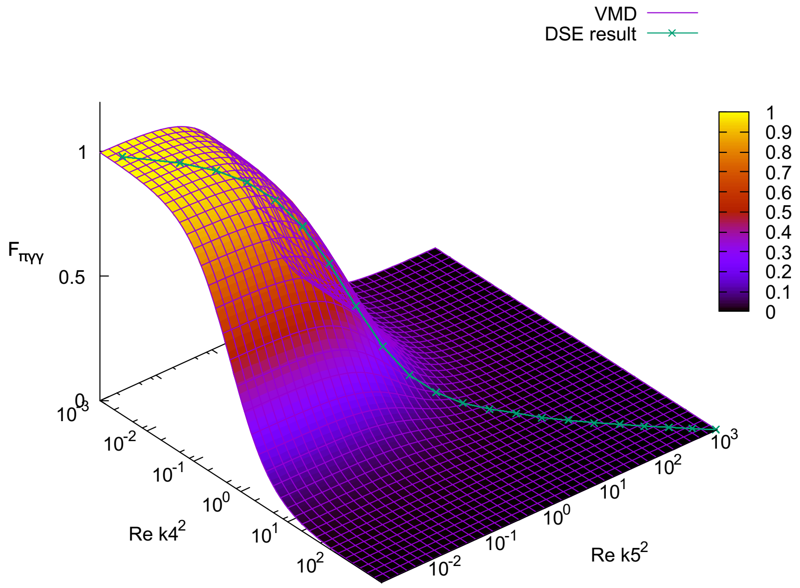


Figure 4.1: Results for the pion form factor plotted against the real part of the two virtual photon momenta. All numbers are in given in GeV. The form factor is compared to the Vector-Meson Dominance model from [26]. Our form factor result is calculated for the symmetric case: $k_{5/4} = 1/2(p \pm P)$ and is represented by the green graph.

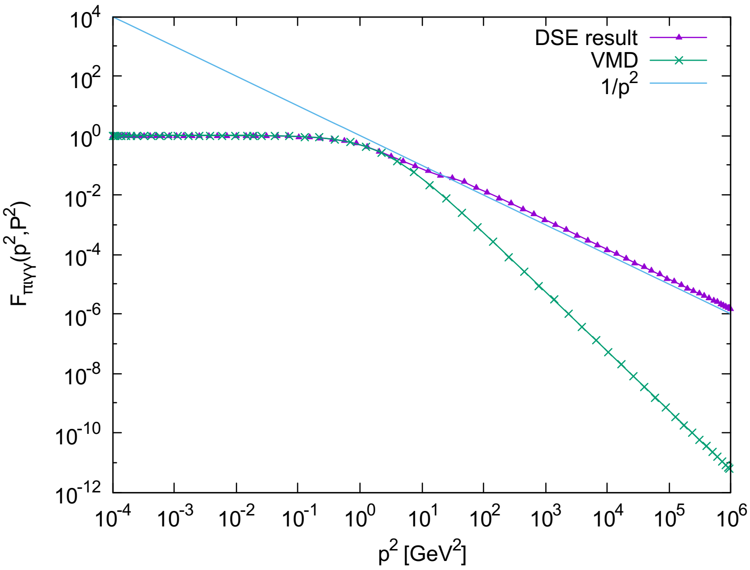


Figure 4.2: Results for the pion form factor plotted against the relative momentum between the photon p on a log-log scale. The form factor is compared to the Vector-Meson Dominance model from [26] and a monopole $1/p^2$.

If we look at Fig.(4.1) we see that our result for the form factor compares well with the VMD. While the VMD is only a model assumption, our result follows directly from QCD's mathematical framework by using the DSE's. Fig.(4.2) shows the form factor on a log-log plot. In this plot we note that the behavior of the form factor in the UV region differs from the VMD model. It falls off like monopole rather than the VMD dipole. In this sense the VMD model is a good approximation only for low p^2 values.

The result show to be consistent with the former DSE calculation [31]. The form factor supposably depends on two variables: The angle between the relative and the total momentum $\hat{Pp}$ and the magnitude of the relative momentum $|p|$. In Fig.(4.2) the form factor is plotted against p and for all different angles. We notice that the angle dependence is negligible. Thus the form factor can be taken to depend on one variable: $F_{\pi\gamma\gamma}(|p|)$.

Therefore it is easy to create a parametrization. Good agreement seems to be achieved with the following parameterization:

$$F_{\pi^0 \to \gamma\gamma}(p^2) = \frac{a}{a + p^2} \quad \text{with} \quad (a = 0.978733 \pm 0.0027)\,\text{GeV}. \tag{4.3}$$

The parameterization has a slight offset in the UV region but show the desired asymptotic behavior. The ansatz for a parameterization will be improved in future work.

Lattice results

Within **lattice QCD** we are able to do a model independent full-error budget calculation of the ($\pi^0 \to \gamma\gamma$) form factor. The result that is discussed here is taken from [19]. In a full-error-budget calculation on the lattice, one must estimate the systematic errors arising from non-zero lattice-spacing, finite-volume effects and unphysical quark masses. One must also incorporate the statistical uncertainties from Monte-Carlo integration. Lattice calculations start with the path-integral and discretize it in terms of a lattice spacing a. In addition to the lattice spacing the calculation has to be performed in finite volume, introducing the volume length L. Finite volume effects constrain the choice of the pion and photon momenta of the decay. We can only choose the pion momentum to be of the form $P = 2\pi\vec{n}/L$ where $\vec{n}$ is any integer three vector. This constraint also applies for one of the photon momenta. The last degree of freedom remaining is the energy of the other photon. Due to these constraints the calculations are only able to extract the off shell form factors on various contours given by choices of $\vec{p}_1$ and $\vec{P}$. In Ref. [19] the authors consider two contours, choosing either the first photon four vector to be $\vec{p_1} = \frac{2\pi}{L}(0,0,1)$ and then $\vec{P} = \frac{2\pi}{L}(0,0,0)$, or the opposite around. The second of these contours comes close to the pion's on shell point, which is desirable since it could be compared with the perturbative calculation and experiment. The result obtained in this paper is consistent with the perturbative result and experimental data for the pion form factor: $F(m_\pi^2, 0, 0) = (4\pi^2 f_\pi)\mathcal{F}_{\pi^0\gamma\gamma}(m_\pi^2, 0, 0) = 1.005(20)(30)$, with the first error being from statistical nature and the second systematic.

The lattice calculation available so far only evaluates the form factor on few contours as discussed in [19] and in this paper only the on-shell value is quoted. The form factor for virtual photons is thus still the subject of current research.

5 More neutral pion decays

With the knowledge of the $\pi^0 \to \gamma\gamma$ form factor we can go on and calculate other pion decays that include this form factor as an intermediate building block. In particular we are interested in the three body decay $\pi^0 \to \gamma e^+ e^-$, also called the Dalitz decay or the 2 body $\pi^0 \to e^+ e^-$ decay, which will be titled 'rare' decay in this thesis. The form factor $\mathcal{F}_{\pi\gamma\gamma}$ includes all non-perturbative information of the decay and thus the remaining calculation, presented here, is perturbative. Therefore we can expand in terms of the QED coupling constant α_e and only consider the diagrams in leading order. The uncertainty due to this approximation can easily be estimated and is small.
To get the final decay rate, for either of the decays, the decay amplitudes need to be squared and spin summed. This way we consider all possible spin states for the outgoing particles. After that we need to integrate over phase space to consider all possible momentum directions of the final state particles. In general the decay rate for a particle A with the energy E_A is given by

$$\Gamma(A \to n) = \frac{S}{2E_A k!} \int \prod \left[\frac{d^3 P_i}{(2\pi)^3 2E_i} \right] (2\pi)^4 \delta^4(P_A - \sum_i P_i) \left[\frac{1}{2J+1} |\mathcal{M}|^2 \right] \tag{5.1}$$

where $\mathcal{M} = \langle P_1, \dots P_n | H | A(P_A) \rangle$ is the spin average and squared decay amplitude, the spin of the particle is denoted by J and the additional factor of $1/k!$, if there are k particles of the same species in the final state.

5.1 The Dalitz Decay: $\pi^0 \to \gamma e^+ e^-$

The three-body, or so called Dalitz decay describes the process of a pion decaying into one photon and an electron and positron ($\pi^0 \to \gamma e^+ e^-$). After the decay into two photons it is the second most likely decay of the neutral pion and has a branching ratio of $(1.174 \pm 0.035)\%$ [32]. The ($\pi^0 \to \gamma\gamma$) form factor is part of this decay since the pion first decays into two photons and one of these photons decays into an electron and positron pair. In this case one of the photons is on-shell and the other is off-shell, therefore we need in particular the $\mathcal{F}_{\pi\gamma\gamma^*}(p, P)$. The decay can be seen as a Feynman diagram in Fig.(5.1)

The branching ratio of a decay is defined via

$$R(\pi^0 \to \gamma e^+ e^-) = \frac{BR(\pi^0 \to \gamma e^+ e^-)}{BR(\pi^0 \to \gamma\gamma)} = C |\mathcal{A}(M_\pi^2)^2| \tag{5.2}$$

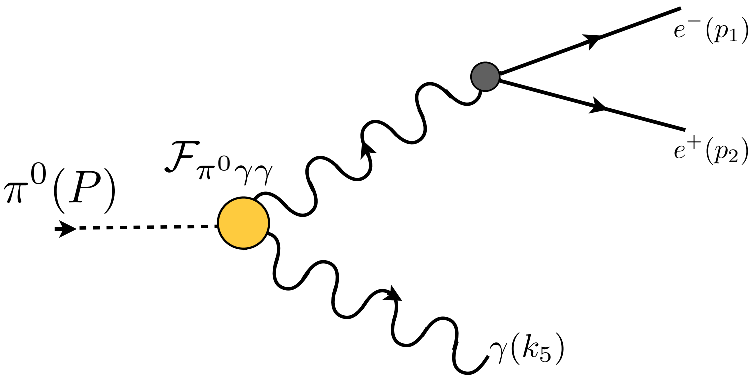


Figure 5.1: The three body decay of the neutral pion into a photon and an electron positron pair. The $\pi \to \gamma\gamma$ form factor is part of the decay and is represented by the yellow dot.

where $\mathcal{A}$ is the decay amplitude, which includes all the dynamic information of the decay. The C upfront is a constant that comes from the integration over phase space which will be determined in course of this section. In particular the decay amplitude comes together as

$$A_{s_1 s_2} = \epsilon_5^\nu \; F_\pi^{\mu\nu}(k_4, k_5) \;\; \frac{-i}{k_4^2} \left(g^{\mu\alpha} - (1-\chi) \frac{k_4^\mu k_4^\alpha}{k_4^2} \right) \; \bar{u}_{s_1}(p_1) \; (ie)\gamma^\alpha \; v_{s_2}(p_2), \tag{5.3}$$

with the polarization vector of the photon in shorthand for $\epsilon_5^\nu = \epsilon_\lambda^\nu(k5)$, the pion form factor $F^{\mu\nu} = \varepsilon^{\mu\nu\alpha\beta} k_4^\alpha k_5^\beta \mathcal{F}_{\pi\gamma\gamma}(P, p)$, which contains the non-perturbative parts of the decay, followed by the perturbative photon propagator $P^{\mu\alpha}(k_4)$ in an arbitrary gauge, which we are free to choose. The factor of ie comes with the electron-photon vertex and the whole expression is multiplied by the spinors for an out-coming positron $v_2 = v_{s_2}(p_2)$ and a out-going electron $\bar{u}_1 = \bar{u}_{s_1}(p_1)$. The indices denoted by greek letters are Lorenz indices.

In the following calculation we fix the QED gauge parameter $\chi = 1$, namely Feynman gauge. To calculate the decay rate $\Gamma_{\pi^0 e^+ e^-}$ we square the amplitude, which means we need to multiply it with its complex conjugated. The complex conjugated decay amplitude equals $A^* = (\epsilon_5^\rho)^* \; \bar{v}_2 \bar{A}_\rho u_1$, with $A_\rho = eF_\pi^{\mu\rho} \; P^{\mu\alpha}(k_4) \; \gamma^\alpha$. Since the amplitude has a rather simple Dirac structure we see immediately that $A_\rho = \bar{A}_\rho$. This corresponds to

$$|\mathcal{A}|^2 = \epsilon_5^\nu (\epsilon_5^\rho)^* (\bar{u}_1 A_\nu v_2)(\bar{v}_2 A_\rho u_1). \tag{5.4}$$

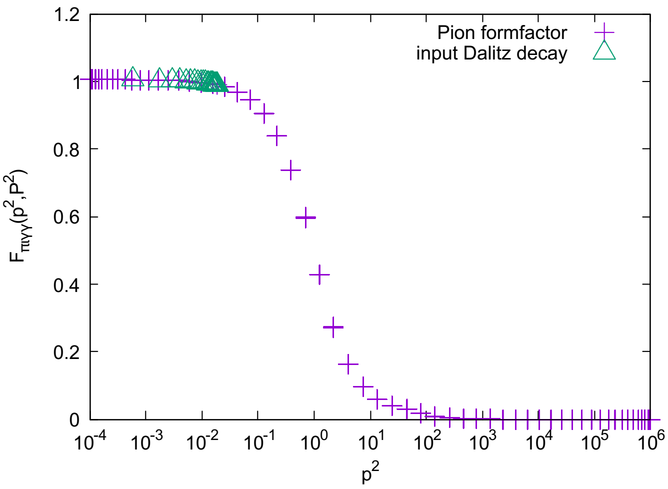


Figure 5.2: Shows the $\mathcal{F}_{\pi\gamma\gamma}$ form factor and the relevant momentum area, that is input to the Dalitz decay. In this area the form factor is close to a constant value of one.

Now we perform the spin sum over the spin of the electron s_2 and the positron s_1, from various quantum field books we know $\sum_{s=\pm} u_s(p)\bar{u}_s(p) = -\not{p}+m$ and $\sum_{s=\pm} v_s(p)\bar{v}_s(p) = -\not{p} - m$ (Ref. [20] ch. 46). Furthermore we do the sum over the polarization vectors of the photons using the relation $\sum_{\lambda=\pm} \epsilon^{\nu}_{\lambda}(k)(\epsilon^{\rho}_{\lambda}(k))^* = g^{\nu\rho}$ valid for $k^2 = 0$, which is true in our case since the final photon is on-shell $k_5^2 = 0$. Hence eq. (5.4) becomes

$$\langle|\mathcal{A}^2|\rangle \quad = \quad \sum_{s_1 s_2} |\mathcal{A}^2| = Tr\left[A_{\nu}(-\not{p}_1 + m_e)A_{\rho}(-\not{p}_2 - m_e)\right]. \tag{5.5}$$

eq. (5.5) is the spin and polarization averaged decay amplitude for the three body Dalitz decay ($\pi^0 \to \gamma e^+ e^-$).
Next we will perform the integration over the phase space. Since we have three final state particle, we will have to integrate out each of their momentum. Which makes in total 9 integrals, but since the momenta are related we can use these constrains to make some of the integrations trivial. The details for the three body phase space integration can be found in the Appendix (7.2.2). After we perform the phase space integration our decay rate is given by eq. (7.22):

$$\Gamma(\pi^0 \to \gamma e^+ e^-) = \frac{1}{2^9\pi^4 M_\pi^2} \int_{4M_\pi^2}^{M_\pi^2} ds \sqrt{1 - \frac{4me^2}{s}} \sqrt{\left(\frac{M_\pi^2 + s}{2M_\pi}\right)^2 - s} \int d\Omega_q \langle|\mathcal{A}^2|\rangle, \tag{5.6}$$

where s is a new relative coordinate which is given by $s = (p_1 + p_2)^2$ and $\int d\Omega_q$ indicates the angular integration for $q = 1/2(p_1 - p_2)$. One integration is trivial and gives a factor of 2π. The particular choice for the four vector of the electron p_1 and positron p_2 are given in the appendix in eq. (7.23) and shows the explicit angular dependence.

Using the form factor results calculated in the previous chapter we obtain the decay rate for the rare decay to be

$$\Gamma(\pi^0 \to \gamma e^+ e^-) = 9.41755\ 10^{-11}\text{GeV} \tag{5.7}$$

Which has a 6% difference to the value that is given in the PDG with 9.0684510^{-11}GeV.

The different decays of the neutral pion have already been discussed years ago as can be read in [33]. In this paper the authors mention, that the effect of the pion form factor $\mathcal{F}_{\pi\gamma\gamma}$ to the Dalitz decay is weak. We confirm this observation and note that the range of the relevant values of the form factor is rather small and at the beginning of the decay function, as displayed in Fig(5.2). The form factor in this area is close to a constant value of one. Therefore the Dalitz estimate of $F_{\pi\gamma\gamma} \sim 1$, first used in [34], is a good estimate for the Dalitz decay and equals the PDG value up to the same discrepancy than the result obtained using the pion form factor.

5.2 The rare decay: $\pi^0 \to e^+e^-$

A less probable decay of the neutral pion is the two body decay of the pion into an electron positron pair via two intermediate photons. In this thesis I titled it the rare decay. The most recent result of this decay were measured from the KTeV collaboration at Fermilab. With a branching ratio of $BR(\pi^0 \to e^+e^-) = 6.64 \times 10^{-8}$ it is marginal smaller than the Dalitz decay. Nevertheless it is even more interesting, different theoretical approaches as well as experimental measurements are interested in the right decay rate. Up to now the results of theory and experiment do not quite agree (up to a 3 sigma discrepancy). That may be be due to the fact that experimental data is not as precise yet, since it is hard to measure a process with such a small probability among other products of the neutral pion decaying. It could also mean that the theoretical description is missing some effects and we have to rethink aspects of our theoretical model or include higher order corrections on our truncation. In any case it is interesting to calculate this decay and within the Dyson-Schwinger it has never been looked at before.

The Dalitz as well as the rare decay have been measured at KTeV-E799 at Fermilab, [35]. The heart of the KTeV facility is a general purpose neutral kaon beam. 800GeV protons hit a BeO target and produce two nearly parallel neutral beams. The rare decay was studied within the ($K \to 3\pi^0$) decay as a source of "tagged" (detected) π^0 decays. The main challenge in such a measurement is to distinguish the signal candidate for the decay of interest. In this case the rare decay was found by normalizing signal candidates of the Dalitz decay. At KTeV they observed almost

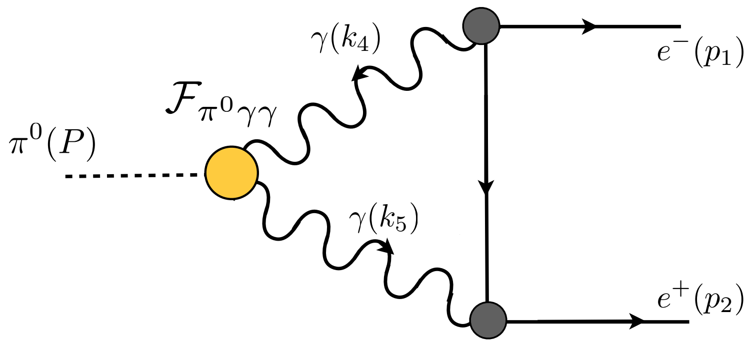


Figure 5.3: The two body decay of the neutral pion into an electron positron pair. The $\pi \to \gamma\gamma$ form factor is part of the decay and is represented by the yellow dot. The vertical line connecting the photon electron vertices is a perturbative electron propagator.

800 events of $\pi \to e^+e^-$ and obtained a precise measurement of the branching ratio: $B(\pi^0 \to e^+e^-, x_D > 0.95) = (6.44 \pm 0.25 \pm 0.22) \times 10^{-8}$, where $x_D = (m_{e^+e^-}/m_{\pi^0})^2$. The first error comes from data statistics alone and the second is the total systematic error. The measurements were found to be consistent with standard model predictions and fall in between predictions of the VMD and χPT.

The rare decay has been calculated using different non-perturbative methods, for example with dispersions relations as can be seen in Ref. [18,36]. In these paper an Ansatz was used for $\mathcal{F}_{\pi^0\gamma\gamma}$, which follows constrains by the data from CLEO and CELLO and restrictions from QCD. The result obtained in the papers were 3σ below the KTeV measurements.

As well as for the Dalitz decay the pion form factor is needed as input. The two decays only differ in the perturbative calculation, since all the non-perturbative information is included in the $\mathcal{F}_{\pi^0\gamma\gamma}$ form factor. The rare decay needs the off-shell result of $F_{\pi\gamma^*\gamma^*}(P,p)$, since the two photons are both virtual.

The decay amplitude is given by

$$A_{s_1s_2}(p_1,p_2) = \int \frac{d^4p}{(2\pi)^4}\ \bar{u}_1\ \left[P^{\mu\alpha}(k_5)\ F^{\mu\nu}_{\pi\gamma\gamma}(P,p)\ P^{\nu\beta}(k_4)\ ie\ \gamma^\beta\ S_e(k_6)\ ie\gamma^\alpha\right]\ v_2. \quad (5.8)$$

with the tensor structure of the pion form factor $F^{\mu\nu}_{\pi\gamma\gamma} = F_{\pi\gamma\gamma}\varepsilon^{\mu\nu}_{k_4k_5}$ the photon propagator $P^{\mu\alpha}$ and the electron propagator $S_e = -i\not{p} + m_e$. Here I introduce the short cut $\varepsilon^{\mu\nu}_{k_4k_5} = \varepsilon^{\alpha\beta\mu\nu}k_4^\alpha k_5^\beta$. The photon propagator are used in Feynman gauge $\chi = 1$, as chosen for the Dalitz decay eq.(5.3). The straight-forward procedure to obtain the spin averaged decay amplitude would be to square $A_{s_1s_2}$ and the sum over all spin configurations. For this decay this would lead us to two four dimensional integrals, and once the product is expressed as a trace it is not obvious how these integrals can be separated. Since evaluating the two integrals together is numerically expensive, I choose to evaluate the integral first and square it afterwards. The two spinors do not depend on the loop momentum p, thus we can perform the integration at the level of the four-dimensional matrix, which combines all the terms within the rectangle brackets. I will denote the matrix as $A(p_1, p_2)$. Within the matrix all Lorenz indices are contracted. After integration we square the amplitude and obtain

$$|\mathcal{A}^2| = (\bar{u}_1 A(p_1,p_2)v_2)\ (\bar{v}_2\bar{A}(p_1,p_2)u_1). \tag{5.9}$$

The complex conjugated object $\bar{A} = \gamma^0 A^\dagger \gamma^0$. Now we use the cyclic properties of the trace and take the spin sum to find

$$\langle|\mathcal{A}^2|\rangle = \sum_{s_1s_2}|\mathcal{A}^2| = Tr\left[A(p_1,p_2)\ (-\not{p}_2 - m_e)\bar{A}(p_1,p_2)\ (-\not{p}_1 + m_e)\right]. \tag{5.10}$$

With the expression for the squared and spin summed decay amplitude we can further proceed to calculate the total decay rate. Therefore we integrate over the two body phase space. Using conservation of total momentum, as well as Lorenz invariance and the fact that all particles are on-shell, we are left with no degrees of freedom. For this reason the phase space integral is trivial and can be found in detail in the appendix (7.2.1). The total decay rate is then given by

$$\Gamma(\pi^0 \to e^+e^-) = \frac{\sqrt{M_\pi^2/4 - m_e^2}}{8\pi M_\pi^2}\langle|\mathcal{A}^2|\rangle. \tag{5.11}$$

The straight-forward implementation of the decay amplitude eq. (5.8) encounters some problems. Performing the loop integration in the decay amplitude we will hit poles. The poles of the two photons and the electron pole are in the integration interval and this makes it impossible to numerically evaluate the amplitude without explicitly extracting the poles. One elegant way to resolve this problem is the usage of tensor decomposition. This will allow us to rewrite the amplitude as a partly analytic expression, which we will be able to partially integrate by hand. The idea for the decompositions follows the ideas for the Nucleons Compton scattering presented in [37].

Tensor decomposion

The tensor structure of the perturbative part in this decay resembles the structure of the diagrams used to describe Compton scattering. The Compton amplitude can

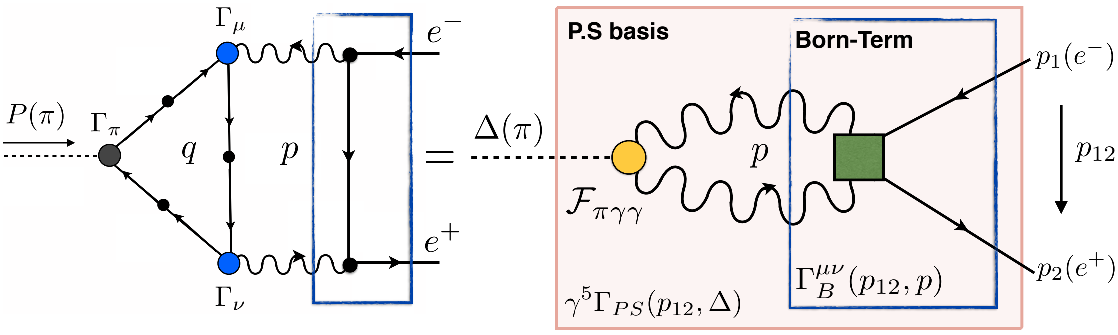


Figure 5.4: Tensor decomposition of the rare decay. The Born-term is indicated by the blue box and represents the electron-photon vertices and the exchanged electron propagator. The red shaded box represents the "vertex" for a pion into an electron positron pair and has the structure of pseudo-scalar basis (P.S).

be decomposed into a Born-term and a one-particle irreducible structure part. The Born-term Γ_B is the simplest interaction between nucleon and photon, a interchanging nucleon propagator. In our case the nucleon is a electron and the nucleon-photon vertices are perturbative electron-photon vertices, which are just bare vertices. Fig.(5.4) shows the diagram for the rare decay and the Born-Term denoted with a blue rectangle. In reference [37] the authors describe that the full Compton amplitude can be decomposed into 18 Lorenz-covariant tensor basis τ_i times their dressing functions c_i. The Born-term has the form:

$$\Gamma_B^{\mu\nu}(p_{12}, p) = \frac{1}{m_e} \frac{1}{4\lambda^2 - \eta_-^2} \sum_{i=1}^{18} c_i \left[\Lambda_+(p_2)\tau_i^{\mu\nu}\Lambda_+(p_1)\right] \tag{5.12}$$

which in our case will be equivalent to $\Gamma_B^{\mu\nu} = ie\ \gamma^\nu\ S_e(k_6)\ ie\gamma^\mu$. The Born-term depends on the relative momentum between the electron and positron p_{12} and between the photons p. The dimensionless scalar variables appearing as pole are given by $\eta_- = (k_4.k_5)/m_e^2$ and $\lambda = (p_{12}.p)/2me^2$, a detailed list of the kinematic variables can be found in the appendix (7.2.1). The positive energy projectors is defined as

$$\Lambda_+(p) = \frac{-ip + m_e}{2m_e} \tag{5.13}$$

with the identities $\Lambda_+(p)u(p) = u(p)$ and $\bar{v}(p)\Lambda_+(p) = \bar{v}(p)$. The tensor structure of the Born-term reduces to only three basis components using bare vertices. Namely $c_1 = 4, c_{10} = 1, c_{11} = 0.5$. The detailed tensor basis can be found in reference [37]

The decay amplitude eq(5.8) is expressed including the Born-term Γ_B as

$$A_{s_1 s_2}(p_1, p_2) = \bar{u}_1 \int \frac{d^4p}{(2\pi)^4} \left[\frac{1}{k_4^2 k_5^2}\, \varepsilon^{\mu\nu}_{k_4 k_5}\, F_{\pi\gamma\gamma}(P, p)\, \Gamma_B^{\mu\nu} \right] v_2. \tag{5.14}$$

Since the tensor structure of the form factor is anti-symmetric $\epsilon^{\mu\nu}_{k_4k_5}$, every symmetric tensor basis in the Born terms vanishes and the total contracted term is given by

$$\varepsilon^{\mu\nu}_{k_4 k_5} \Gamma_B^{\mu\nu}(p_{12}, p) = \frac{1}{m_e} \frac{1}{4\lambda^2 - \eta_-^2} \frac{\eta_-}{i m_e} \Lambda_+(p_2) \{\gamma_5 [\not{\Sigma}, \not{\Delta}], \not{\Sigma}\} \Lambda_+(p_1), \tag{5.15}$$

with the total momentum of the pion $P = \Delta$ and the relative momentum between the internal photons $p/2 = \Sigma$. The anti-commutator has the same tensor basis elements as a pseudo-scalar. We remember the definition of the pseudo-scalar basis: $\gamma_5 \Gamma_{PS}(p_{1,2}, \Delta)$,with $\Gamma_{PS} \in \{\mathbb{I}, \not{p}_{12}, \not{\Delta}, [\not{p}_{12}, \not{\Delta}]\}$, as first introduced in section (3.2). Projecting onto this basis and using relation between the various kinematic variables defined in section. (7.1.1), is rewritten to

$$A_{s_1 s_2}(p_1, p_2) = \bar{u}_1 \left(-4i\Lambda_+^2 \gamma_5 \left[\int_\Sigma \frac{1}{\eta_+^2 - \omega^2} \frac{\eta_-}{4\lambda^2 - \eta_-^2} (\omega \not{\Sigma} - \sigma \not{\Delta}) F_{\pi\gamma\gamma} \right] \right) v_2. \tag{5.16}$$

Here the photons pole are rewritten into the dimensionless kinematic variables $1/(k_4^2 k_5^2) = 1/(\eta_+^2 - \omega^2)$, the integration variable is the loop momentum $\Sigma = 1/2\ p$ and $\int_\Sigma$ indicates the four dimensional integration.

The decay process itself ($\pi^0 \to e^+ e^-$) has pseudo-scalar quantum numbers as well. Thus we can describe the shaded rectangle in Fig.(5.4) using the pseudo-scalar basis. Due to symmetry the first and the fourth basis can directly shown to disappear. This leave the second and the third basis element $\tau_i \in \{\not{p}_{12}, \not{\Delta}\}$. Using the properties of the positiv energy projectors we can conclude that the third component will cancel out to be zero and we are left with the second $\tau_3 = \not{\Delta}$:

$$\begin{aligned}
\Lambda_+^2 \left(\gamma_5\ \not{p}_{12} \right) \Lambda_+^1 \quad &= \quad \Lambda_+^2 \left(-\frac{\not{p}_2}{2} \gamma_5 + \gamma_5 \frac{\not{p}_1}{2} \right) \Lambda_+^1 = 0 \\
\Lambda_+^2 \left(\gamma_5\ \not{\Delta} \right) \Lambda_+^1 \quad &= \quad \Lambda_+^2 \left(-\not{p}_2 \gamma_5 - \gamma_5 \not{p}_1 \right) \Lambda_+^1 = -2i m_e \Lambda_+^2 \gamma_5 \Lambda_+^1.
\end{aligned} \tag{5.17}$$

The dressing function of this component is calculated by projecting Γ_{PS} with τ_3 and taking the trace.

$$\frac{1}{4} Tr \left[\not{\Delta} \Gamma_{PS} \right] = f_3\ \Delta^2 = m_e^2 \int_\Sigma \frac{1}{\eta_+^2 - \omega^2} \frac{\eta_-}{4\lambda^2 - \eta_-^2} \left(\omega^2 - 4\sigma t \right) F_{\pi\gamma\gamma}(\Delta, \Sigma) \tag{5.18}$$

Therefore the dressing function f_3 is calculated via

$$\begin{aligned}
f_3 \quad &= \quad \frac{1}{(2\pi)^3} \frac{1}{2} \int d\sigma \sigma \int_{-1}^{1} dz \sqrt{1 - z^2}\ F_{\pi\gamma\gamma}(\sigma, z) \frac{\eta_-}{\eta_+^2 - \omega^2} \left(\frac{\omega^2}{4t} - \sigma \right) \\
&\times \quad \int_{-1}^{1} dy \frac{1}{4\lambda^2 - \eta_-^2},
\end{aligned} \tag{5.19}$$

where $\lambda(y)$ depends on y and the integration variable is substituted to a dimensionless variable $\sigma = \Sigma^2/m_e^2$. One of the angular integration is simple and give a factor of 2π. Using the tensor decomposition we were able to rewrite the integrand and now we have a explicit analytic form for the y-dependence.

$$
\begin{aligned}
f_3 &= \frac{1}{(2\pi)^3}\frac{1}{2}\int d\sigma\sigma \int_{-1}^{1} dz\sqrt{1-z^2}\; F_{\pi\gamma\gamma}(\sigma,z)\frac{\sigma(1-z^2)}{(\sigma-t)^2+4\sigma t(1-z^2)} && (5.20)\\
&\times \int_{-1}^{1} dy \frac{1}{(\sigma-t)^2+4\sigma(1+t)(1-z^2)y^2}, && (5.21)
\end{aligned}
$$

with the dimensionless kinematic variables $t = \Delta^2/(4m_e^2)$. The pion form factor is the only non-analytic structure in the integrand, but does not depend on y. Thus we can perform the y integration analytically. The decay amplitude expressed in terms of a pseudo-scalar basis is given by

$$
A_{s_1s_2}(p_1,p_2) = \bar{u}_1\left(-8m_e\;\Lambda_+^2\;\gamma_5 f_3\;\Lambda_+^1\right)v_2. \tag{5.22}
$$

Results

Unfortunately up to this point I have not been able to calculate the decay rate for ($\pi^0 \to e^+e^-$). With the tensor decomposition explained in the paragraph above the calculation comes down to a two dimensional integral that can be calculated using the results for the form factor $\mathcal{F}_{\pi\gamma^*\gamma^*}$. Attempts suggest that the influence of the form factor for the decay rate is small. But the asymptotic behavior in the UV region is an important feature to ensure the convergence of the integral.

6 Conclusion and Outlook

In this thesis we calculated neutral pion decays using the non-perturbative Dyson-Schwinger-Bethe-Salpeter formalism. In particular the electromagnetic form factor of the neutral pion $\mathcal{F}_{\pi^0\gamma\gamma}$, the three body Dalitz decay ($\pi^0 \to \gamma e^+ e^-$) were calculated and attempts for the less frequent ($\pi^0 \to e^+ e^-$) decay were made.

The form factor was calculated for the on-shell quantities and compared to experimental results and the values obtained by the $SU(3)$ axial anomaly in perturbation theory and found to be in 2% agreement. For off-shell photons the form factor was compared to the VMD model [26] and showed to be in agreement for low momenta. The result of former DSE studies [31] could be confirmed and it was noticed that the form factor shows no angular dependence and thus is fully determined by the absolute value of the relative momentum: $\mathcal{F}_{\pi^0\gamma\gamma}(p)$.

The decay rate for the Dalitz decay calculated using the numerical result for $\mathcal{F}_{\pi^0\gamma\gamma}$ was found to be within 10% of the experimental value. Comparing to a calculation with a constant value as input for $\mathcal{F}_{\pi^0\gamma\gamma} = 1$ we found that the results appeared to have roughly the same error. This observation was logical since the values relevant for the decay cover a small momentum region in the low energy range of $\mathcal{F}_{\pi^0\gamma\gamma}$. In this region the form factor is well approximated to be constant.

For the rare decay we discovered that the relevant diagrams have the same basic structure as the Compton amplitude, describing the resonances of the nucleon. But in the case of the decay considered here the electron and positron couple only electromagnetically to the photons, which reduces the structure to bare perturbative vertices. The decomposition introduced in [37] reduces significantly and the expression for the decay rate breaks down to a simple integral equation. Attempts to calculate the rate seem to suggest that the influence of the form factor is weak, but the asymptotic behavior is important for the convergence of the integral. The calculation will be pursued further in upcoming work.

With the results of $\mathcal{F}_{\pi^0\gamma\gamma}$ we obtained an important input for different QCD processes, such as g-2, and we could show that within the Dyson-Schwinger formalism using the rainbow-ladder truncation we obtain good results for neutral pion decays. Possible future work includes extending the calculation to the decay of other pseudoscalar particles like the η. Indeed there is much interesting work to be done in the sector of meson decay in the DSE formalism.

7 Appendix

7.1 momentum set-up and coordinates

In the calculation of the $(\pi^0 \to \gamma\gamma)$ from factor, spherical coordinates were used for the loop integration, thus the momenta were chosen the following

$$
\begin{aligned}
q =& |q| \begin{pmatrix} \sin\psi\sin\theta\sin\varphi \\ \sin\psi\sin\theta\cos\varphi \\ \sin\psi\cos\theta \\ \cos\psi \end{pmatrix} = |q| \begin{pmatrix} \sin\varphi\sqrt{1-y^2}\sqrt{1-z12} \\ \cos\varphi\sqrt{1-y^2}\sqrt{1-z^2} \\ y\sqrt{1-z^2} \\ z \end{pmatrix} \\
p =& |p| \begin{pmatrix} 0 \\ 0 \\ y_p\sqrt{1-z_p^2} \\ z_p \end{pmatrix} \quad , \quad P = \begin{pmatrix} 0 \\ 0 \\ 0 \\ iM_\pi \end{pmatrix} ,
\end{aligned}
\tag{7.1}
$$

with the relative momentum between the photons p, the total momentum P and the loop momentum q.The momenta given in the form factor diagram Fig. (3.7) are related through

$$
\begin{aligned}
k_1 &= p + \sigma P & k_2 &= p - (1-\sigma)P \\
k_4 &= \frac{1}{2}(p - P) & k_5 &= \frac{1}{2}(p + P) ,
\end{aligned}
\tag{7.2}
$$

with the momentum distribution parameter $\sigma \in \{0, 1\}$.

7.1.1 rare decay

For the tensor decomposition of the rare decay a number of different relative momenta were introduced to match the conventions used in [37]:

$$
\begin{aligned}
p_{12} &= \frac{1}{2}(p_1 + p_2) & \Delta &= k_5 - k_4 = p2 - p1 = P \\
\Sigma &= \frac{1}{2}(k_4 + k_5) = \frac{1}{2}p
\end{aligned}
\tag{7.3}
$$

with the photon momenta k_4, k_5, the electron p_1, positron p_2, the total momentum P, the relative momentum between photons p and the relative momentum between electron and positron p_{12}. Check out Fig.(5.4) for an overview. The inverse relations follow

$$p_1 = p_{12} - \frac{\Delta}{2} \qquad\qquad k_5 = \Sigma + \frac{\Delta}{2} \tag{7.4}$$
$$p_2 = p_{12} + \frac{\Delta}{2} \qquad\qquad k_4 = \Sigma - \frac{\Delta}{2}$$

Furthermore the following dimensionless variables are introduced

$$\eta_+ = \frac{k_4^2 + k_5^2}{2m_e^2} \qquad \eta_- = \frac{k_4 \cdot k_5}{m_e^2} \qquad \omega = \frac{k_5^2 - k_4^2}{2m_e^2} \tag{7.5}$$
$$\lambda = \frac{p_{12} \cdot \sigma}{m_e^2} = \frac{p_{12} \cdot k_5}{m_e^2} = \frac{p_{12} \cdot k_4}{m_e^2} \qquad t = \frac{\Delta}{4m_e^2} = \frac{\eta_+ + \eta_-}{2} \qquad \sigma = \frac{\Sigma^2}{m_e^2}.$$

7.2 Phase space integrations

In this paragraph we present the phase space integration used for the three-body Dalitz ($\pi^0 \to \gamma e^+ e^-$) and the two-body rare decay ($\pi^0 \to e^+ e^-$). The evaluation for the phase space integral follows the lecture notes of Martin Savage (University of Washington) [38].

7.2.1 Two particle phase space integration

Due to rotational invariance we can choose the three momentum of the out-coming particle to point along the z-axis, which leaves us with a multiplicative factor for the whole phase space integration.
The decay rate for the rare decay ($\pi^0 \to e^+ e^-$) is calculated via

$$\Gamma(\pi^0 \to e^+ e^-) = \frac{1}{2E_\pi} \int \frac{d^3 p_1}{(2\pi)^3 2E_1} \int \frac{d^3 p_2}{(2\pi)^3 2E_2} (2\pi)^4 \delta^4(P - p_1 - p_2) \langle |\mathcal{A}^2| \rangle, \tag{7.6}$$

where $\langle |\mathcal{A}^2| \rangle$ is the spin averaged squared decay amplitude as given in eq. (5.10), p_1 and p_2 are the momenta of the out-coming electron and positron and P the total momentum of the pion. Since the decay amplitude is Lorenz invariant we choose to work in the rest frame of the pion, such that P is chosen as eq. (7.1). In the rest frame the electron and positron have the same energy $E_{1,2} = \sqrt{(\vec{p}_{1,2})^2 + m_e^2}$, since $\vec{p}_1 = -\vec{p}_2$. First the integral over p_1 will be evaluated. The delta function can be rewritten using $\int d^4 p_1\, \theta(E_1) \delta(E_1^2 - me^2) = \int d^3 p_1/(2E_1)$, thus the integral is trivial and we are left with an one dimensional delta function

$$\int \frac{d^3 p_1}{2E_1} \delta^4(P - p_1 - p_2) = \delta((P - p_2)^2 - m_e^2). \tag{7.7}$$

For the integral over the second momentum p_2 we evaluate the argument of the delta function: $\delta((P - p_2)^2 - m_e^2) = \delta(M_\pi^2 - 2M_\pi E_2) = \delta(M_\pi - 2E_{1,2})/(2E_1)$. Using spherical coordinates, the three dimensional integral can be decomposed as the integral over two angles and one magnitude $\int d^3p_2/(2E_2) = 1/2 \int \; d\Omega_2 \; dp_2 \; p_2^2/E_2$. The integration over the angles is trivial and gives a factor of 4π in front, since the decay amplitude does not depend on the orientation of the momenta. The argument of the delta function can be further rewritten to

$$\begin{aligned}
\delta(M_\pi - 2\sqrt{M_\pi^2/4 - m_e^2}) &= \delta(p_1\sqrt{M_\pi^2/4 - m_e^2}) \left| \left(\frac{\partial}{\partial p_2} 2\sqrt{p_2^2 + m_e^2} \right) \right|^{-1} \\
&= \frac{E_2}{2\, p_2} \, \delta(p_2 - q), \qquad (7.8)
\end{aligned}$$

with $q = \sqrt{M_\pi^2/4 - m_e^2}$. The integration over p_2 evaluates the delta function and we end up with:

$$\begin{aligned}
\Gamma(\pi^0 \to e^+e^-) &= \frac{1}{8\pi^2 E_\pi} \frac{1}{(2E_{1,2})^2} \, 4\pi \int dp_2 \; p_2^2 \, \frac{E_2}{2\, p_2} \delta(p_2 - q) \, \langle |\mathcal{A}^2| \rangle \qquad (7.9) \\
&= \frac{q}{8\pi M_\pi^2} \, \langle |\mathcal{A}^2| \rangle. \qquad (7.10)
\end{aligned}$$

The phase space integral reduces to a multiplicative factor in front of the decay amplitude $\langle |\mathcal{A}^2| \rangle$. Due to rotational invariance we are free to choose the direction of the outgoing particle and we take it to be along the z-axis. With this choice and the restrictions due to the on-shellness of the particles, we define the four-vector of the outgoing electron and positron to be

$$p_1 = \begin{pmatrix} 0 \\ 0 \\ \sqrt{M_\pi^2/4 - m_e^2} \\ -iM_\pi/2 \end{pmatrix} \quad , \quad p_2 = \begin{pmatrix} 0 \\ 0 \\ \sqrt{M_\pi^2/4 - m_e^2} \\ iM_\pi/2 \end{pmatrix}. \qquad (7.11)$$

7.2.2 Three particle phase space integration

The decay rate for the three body or Dalitz decay is given by

$$\begin{aligned}
\Gamma(\pi^0 \to \gamma e^+e^-) &= \frac{1}{2M_\pi} \int \frac{d^3k_5}{(2\pi)^3 2E_5} \int \frac{d^3p_1}{(2\pi)^3 2E_1} \int \frac{d^3p_2}{(2\pi)^3 2E_2} \qquad (7.12) \\
&\times \; (2\pi)^4 \delta^4 \left(P - k_5 - p_1 - p_2 \right) \langle |\mathcal{A}^2| \rangle,
\end{aligned}$$

with the spin averaged and squared decay amplitude $\langle |\mathcal{A}^2| \rangle$ as given in eq. (5.5) and the phase space integrals over all three final state particle. The denominator for each integral comes from the definition of the phase space integral and the four dimensional delta function comes with a factor of $(2\pi)^4$. We use a the same trick as in the two particle phase space integration and rewrite the electron and positron integration, such that

$$\int \frac{d^3p_1}{2E_1} \int \frac{d^3p_2}{2E_2} = \int d^4p_1 \int d^4p_2 \; \theta(p_1^0)\theta(p_2^0)\delta(p_1^2 - m_e^2)\delta(p_2^2 - m_e^2) \tag{7.13}$$

Going forward it makes sense to do a change of variables to these orthogonal coordinates

$$k = p_1 + p_2 = \sqrt{s} \quad , \quad q = \frac{1}{2}(p_1 - p_2) \quad , \quad p_1 = \frac{1}{2}k + q \quad , \quad p_2 = \frac{1}{2}k - q, \tag{7.14}$$

this will make things easier later on. Therefore eq. (7.13) becomes

$$\begin{aligned} \int \frac{d^3p_1}{2E_1} \int \frac{d^3p_2}{2E_2} &= \int d^4k \; d^4q \; \theta\left(\frac{k^0}{2} - q^0\right) \theta\left(\frac{k^0}{2} + q^0\right) \\ &\quad \times \delta\left(\frac{k^2}{4} + q^2 + m_e^2 + k.q\right) \delta\left(\frac{k^2}{4} + q^2 + m_e^2 - k.q\right) \\ &= \frac{1}{2} \int d^4k \; d^4q \; \theta\left(\frac{k^0}{2} \pm q^0\right) \delta\left(\frac{k^2}{4} + q^2 - me^2\right) \delta\left(k.q\right) \int ds \; \delta(s - k^2). \end{aligned} \tag{7.15}$$

We inserted a one in the form $\int ds\delta(s - k^2) = 1$ and rewrite the delta functions using the fact that $\delta(a + b)\delta(a - b) = \delta(2b)\delta(a) = 1/2\delta(b)\delta(a)$. For the next steps we will focus one part of the integral. We will use that fact that this part is Lorenz-invariant to evaluate it in the pion rest frame:

$$I = \int d^4q \; \theta(q^0)\theta(k^0) \; \delta\left(\frac{k^2}{4} + q^2 - m_e^2\right) \delta(k.q) \; \langle|\mathcal{A}^2|\rangle. \tag{7.16}$$

In the pion rest-frame $k = (0, 0, 0, i\sqrt{s})$ and $k.q = q^0\sqrt{s}$. The integration over the energy component of q leaves us with

$$I = \theta(k^0) \; \frac{1}{\sqrt{s}} \; \int d^3q \; \delta\left(\frac{s}{4} + |q|^2 - m_e^2\right) \langle|\mathcal{A}^2|\rangle. \tag{7.17}$$

The three dimensional integration can be decomposed into two angular integrals $\int d\Omega_q$and one radial integral over the absolute value of p. First we evaluate the radial integral using the delta function, which is rearranged such that eq(7.17) becomes:

$$\begin{aligned} I &= \theta(k^0) \; \frac{1}{\sqrt{s}} \; \int d\Omega_q \int dq \; |q^2| \frac{1}{2|q|} \; \delta\left(|q| - \sqrt{\frac{s}{4} - m_e^2}\right) \langle|\mathcal{A}^2|\rangle \\ &= \frac{1}{2}\theta(k^0) \; \frac{1}{\sqrt{s}} \sqrt{\frac{s}{4} - m_e^2} \int d\Omega_q \langle|\mathcal{A}^2|\rangle. \end{aligned} \tag{7.18}$$

The angular integration which is left over, is performed in the pion-rest-frame. Now we can proceed and put the expression for I back into eq. (7.15) and conclude to

$$\begin{aligned}
\Gamma(\pi^0 \to \gamma e^+ e^-) &= \frac{1}{4M_\pi(2\pi)^5} \int \frac{d^3k_5}{2E_\gamma} \int d^4k \int ds\ \delta(s-k^2)\delta^4(P-k_5-k) I \\
&= \frac{1}{4M_\pi(2\pi)^5} \int d^4k \int ds\ \delta(s-k^2)\delta\left((P-k)^2 - m_\gamma^2\right) I \\
&= \frac{1}{4M_\pi(2\pi)^5} \int \frac{d^3k}{2E_k} \int ds\ \delta\left(P^2+k^2-2Pk\right) \\
&\times\ \frac{1}{4}\sqrt{1-\frac{4m_e^2}{s}} \int d\Omega_q \langle|\mathcal{A}^2|\rangle. \qquad (7.19)
\end{aligned}$$

We keep using the same trick rewriting the integrals from three dimensional into four dimensional integrals with heavy-side and delta functions and the other way around. From the first to the second equal sign we evaluated the integral over k_5 using the four dimensional delta function.

$$\begin{aligned}
\int \frac{d^3k}{2E_k}\delta(M_\pi^2+s-2M_\pi k^0) &= 2\pi \int dk^0|k|\delta(M_\pi^2+s-2M_\pi k^0) \qquad (7.20)\\
&= \frac{\pi}{M_\pi}|k^*|,
\end{aligned}$$

where the star indicated that $|k|$ is evaluated at the argument of the delta function, which means in detail

$$|k^*| = \sqrt{\left(\frac{M_\pi^2+s}{2M_\pi}\right)^2 - s} \qquad (7.21)$$

Now we evaluated all but one integral. The minimum value that s can reach corresponds to producing the electron positron pair at the threshold and hence the minimal value is $s_{min} = 4M_\pi^2$. The maximal value of s is obtained the photon is produced in the rest frame of the pion, $s_{max} = M_\pi^2$. The final form for the decay rate is given by

$$\Gamma(\pi^0 \to \gamma e^+ e^-) = \frac{1}{2^9\pi^4 M_\pi^2} \int_{4M_\pi^2}^{M_\pi^2} ds \sqrt{1-\frac{4m_e^2}{s}} \sqrt{\left(\frac{M_\pi^2+s}{2M_\pi}\right)^2 - s} \int d\Omega_q \langle|\mathcal{A}^2|\rangle. \qquad (7.22)$$

The difficulty of this decay, in my opinion, lies in the momentum routing. The out-coming particle are on the mass-shell and that constrains the momentum choice in the system. The four vectors for the momenta of the out-coming particle is chosen accordingly

$$p_1 = \begin{pmatrix} 0 \\ \sqrt{\left(\frac{\sqrt{s}}{2}\right)^2 - m_e^2}\sin\theta \\ \sqrt{\left(\frac{\sqrt{s}}{2}\right)^2 - m_e^2}\cos\theta \\ \frac{\sqrt{s}}{2}i \end{pmatrix} \quad , \quad p_2 = \begin{pmatrix} 0 \\ -\sqrt{\left(\frac{\sqrt{s}}{2}\right)^2 - m_e^2}\sin\theta \\ -\sqrt{\left(\frac{\sqrt{s}}{2}\right)^2 - m_e^2}\cos\theta \\ \frac{\sqrt{s}}{2}i \end{pmatrix} . \tag{7.23}$$

I want to note that one might be inclined calculate this phase-space integral in the limit of a massless electrons $m_e \approx 0$, since compared to the pion the electron mass is rather small $m_\pi >> m_e$. Doing so turned out to produce poles in the integrand, which makes it really hard to evaluate the integral harder numerically. Therefore I decided to take a step back and do the phase space integral for all the particles massive and this solved the problem.

7.3 Field theory details

The theory introduction was aimed to be succinct with not to many details but still able to relate. But of course there are more details that could be explained and may help with understanding, this is the reason for this appendix. Of course this goes back the basics field theory lectures and can be found in a more organized way in good books such as [20, 21].

7.3.1 functional derivatives, expectation values & the generating functionals

Functional derivatives play an important role in quantum field theory and as we mentioned in section (2.1), taking the functional derivative with respect to a source J gives an expectation values of the field related to that source. We will start with an example to introduce the reader how to take a functional derivative, let's define the variation partial derivative:

$$\frac{\delta J(x)}{\delta J(y)} = \delta^4(x-y), \tag{7.24}$$

since $J(y) = \int d^4x \delta(x-y) J(x)$. Let me define an example function, which will be similar to the source term in the QCD action, with $F[J] = \int_y \phi(y) \cdot J$ and differentiating $\exp(F[J])$ with respect to J:

$$\begin{aligned} \frac{\delta}{\delta J(x)} e^{F[J]} &= e^{F[J]} \frac{\delta}{\delta J(x)} \int_y \phi \cdot J(y) \\ &= e^{F[J]} \int_y \phi(y) \delta^4(x-y) = e^{F[J]} \phi(x) \ . \end{aligned} \tag{7.25}$$

For each derivative with respect to a source J we get the field $\phi(x)$ connected to this source down from the exponential. Taking higher order derivatives will therefore bring down more fields and produce different expectation values.

The quantum expectation value for a general function $\langle f(\phi)\rangle$ can be rewritten in terms of functional derivatives with respect to the source J:

$$\begin{aligned} \left\langle f(\hat{\phi})\right\rangle = \langle 0|T\ f(\phi)\ |0\rangle &= \frac{\int \mathcal{D}\phi\ e^{-S[\phi]+\int_x \phi(x)J} f(\phi)}{\int \mathcal{D}\phi\ e^{-S[\phi]+\int_x \phi(x)J}} \\ &= \frac{1}{Z[J]} \int \mathcal{D}\phi\ e^{-S[\phi]+\int_x \phi(x)J} f(\phi) \\ &= Z[J]^{-1} f\left(\frac{\delta}{\delta J}\right) Z[J] \end{aligned} \tag{7.26}$$

In the last line of eq. (7.26), the functional derivative is the argument of the general function f. As seen in the example above (eq. (7.25)) each derivative of Z with respect to J will bring down a field and this leads to the function $f(\phi)$ in connection with the path-integral, which is the expectation value $\langle f(\phi)\rangle$. Using the identity

$$e^{-A} f(\partial) e^{A} = f(\partial + \partial A)\mathbf{1}. \tag{7.27}$$

we can further rewriting eq. (7.26) with the specific choice of

$$e^{A} = Z[J] = e^{iW[J]} \quad , \quad \partial A = \frac{\delta}{\delta J}(iW[J]) \ . \tag{7.28}$$

This results in

$$Z[J]^{-1} f\left(\frac{\delta}{\delta J}\right) Z[J] = f\left(\frac{\delta}{\delta J} + i\frac{\delta W[J]}{\delta J}\right) = \langle f(\phi)\rangle_J \, . \tag{7.29}$$

$W[J]$ appearing in eq. (7.28) is the functional for connected Greens function introduced in section(2.1), which is related to the generating functional $Z[J]$ through the definition $W[J] = -i\ \log(Z[J])$. If the general function equals the functional derivative with respect to the fields of the action, $f = \delta S[\phi]/\delta\phi$, eq. (7.29) leads to eq (2.15):

$$\left\langle \frac{\delta S[\phi]}{\delta \phi(x)} \right\rangle = Z[J]^{-1}\ \frac{\delta S}{\delta \phi}\left[\frac{\delta}{\delta J(x)}\right] Z[J] = J(x). \tag{7.30}$$

By taking the functional derivative of different functionals we get different Greens functions. Using the generating functional $Z[J]$ we end up with the full **n-point** Greens functions. Differentiating $W[J]$ leads to all **connected** Greens functions. You can think about this in terms of a simple example: for $Z[J_1, J_2]$ depending on two fields, and thus also two sources we calculate

$$\frac{\delta}{\delta J_1}\frac{\delta}{\delta J_2}\log Z[J] = \frac{\delta}{\delta J_1}\left(\frac{1}{Z}\frac{\delta}{\delta J_2}Z\right) \tag{7.31}$$

$$= \frac{1}{Z}\frac{\delta}{\delta J_1}\frac{\delta}{\delta J_2}Z - \left(\frac{1}{Z}\frac{\delta}{\delta J_1}Z\right)\left(\frac{1}{Z}\frac{\delta}{\delta J_2}Z\right) \tag{7.32}$$

$$= \langle\Phi_1\Phi_2\rangle - \langle\Phi_1\rangle\langle\Phi_2\rangle = \langle\Phi_1\Phi_2\rangle_{connected} \tag{7.33}$$

which gives the expectation value of $\Phi_1\Phi_2$ for all connected Greens functions.
Legendre transforming $W[J]$ leads to another functional $\Gamma[\tilde{\phi}]$ called the effective action, which is the generating functional for the one-particle-irreducible (1PI) Greens functions. The Legendre transformation of $Z[J]$ to $\Gamma[\tilde{\phi}]$ is defined through

$$Z[J] = e^{i(\Gamma[\tilde{\phi}] - \int_x \tilde{\phi} J(x))}. \tag{7.34}$$

The argument in the effective action is the averaged field $\tilde{\phi}$. The averaged field is the vacuum expectation value of ϕ in the background of a source J:

$$\tilde{\phi}(x) = -\frac{\delta W[J]}{\delta J(x)} = \frac{1}{Z[J]}\frac{\delta}{\delta J(x)}Z[J] = \langle\phi(x)\rangle\,. \tag{7.35}$$

A useful notation going on will be writing $W_{xy}[J]'' = \frac{\delta^2 W}{\delta J_x \delta J_y}$ for the second derivative of W and so on. Looking at eq. (7.35) and eq. (7.34) we observe that the source $J(x)$ and the field $\tilde{\phi}(x)$ are related through

$$W_x[J]' = -\tilde{\phi}(x) \quad , \quad \Gamma_x[\tilde{\phi}]' = J(x). \tag{7.36}$$

We note that through these relation we find that the second derivative of the effective action and $W_{xy}[J]''$ are related through:

$$\int_y W_{xy}[J]''\Gamma''_{zy}[\tilde{\phi}] = \int_y \frac{\delta\tilde{\phi(x)}}{\delta J(y)}\frac{\delta J(y)}{\delta\tilde{\phi(z)}} = \frac{\delta\tilde{\phi(x)}}{\delta\tilde{\phi(z)}} = -\delta^4(x-z). \tag{7.37}$$

Since the 1-PI two point function in presence of an external source, better known as the propagator, is an important quantity for further discussion we define it via $\Gamma_{xy}[\tilde{\phi}]'' := \Delta_{xy}[\tilde{\phi}]$. Rewriting the functional derivative by inserting a one using the indentity in eq. (7.37) we obtain

$$\frac{\delta}{\delta J(x)} = \int_y \frac{\delta\tilde{\phi}(y)}{\delta J(x)}\frac{\delta}{\delta\tilde{\phi}(y)} = \int_y \Delta_{xy}[\tilde{\phi}]\frac{\delta}{\delta\tilde{\phi}(y)}. \tag{7.38}$$

With eq. (7.38) and eq. (7.36) we can now rewrite eq. (7.29) and obtain:

$$\langle f(\phi)\rangle_J = f\left(-\frac{\delta W[J]}{\delta J(x)} + \frac{\delta}{\delta J}\right) = f\left(\tilde{\phi}(x) + \int_y \Delta_{xy}[\tilde{\phi}]\frac{\delta}{\delta\phi(y)}\right). \tag{7.39}$$

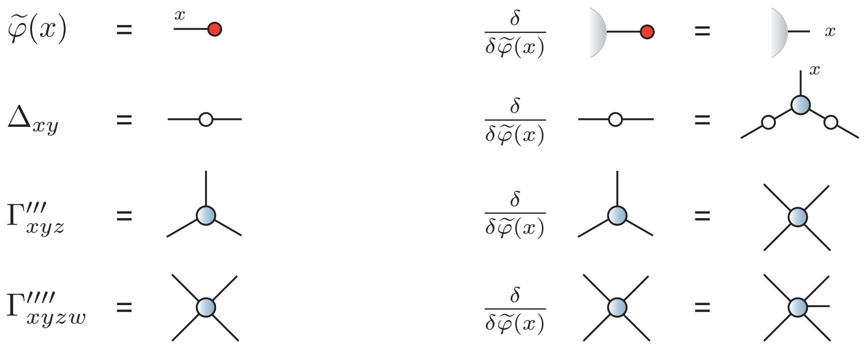


Figure 7.1: Graphical notation of the field and functionals. The red dot indicates a source J and the white dot a fully dressed propagator. All following pictures in this chapter are taking out of the lecture notes from [4].

This was to goal of this section. To reach an expression for the expectation value in terms of the effective action Γ, which can be seen implicit in eq(7.39) in form of the propagator Δ_{xy}. eq. (7.39) defines the expectation value in a quantum field theory. The second term in the bracket represents the deviation of the averaged field due to quantum fluctuations.

7.3.2 Graphical derivation of the DSE

In this section we want to derive the Dyson-Schwinger equation using a graphical description as it was taught in [4]. The starting point for this is the quantum expectation value in eq. (7.39) of the last section.

For the derivation it is more convenient to think about of the equations introduced in section(7.3.1) in terms of diagrams, namely *Feynman diagrams*. This safe us from writing out a great amount of equations. The graphical notation of the fields and functionals can be seen in FIG.(7.1).

A propagator $\Delta(x-y)$ is shown as a connected line with a white dot in the middle, the dot indicates that the propagator is fully dresses. Fully dressed means that the propagator is equal to the sum of all diagrams that have two-external legs. Any loop that can be put on the line gets included in this definition. A simple line with a red dot at the end represents a average field $\tilde{\phi}$ in presence of a source $J(y)$, which is indicated by the red circle at the ends. Taking a functional derivative diagrammatically is represented by removing the source dots. Our aim is to express the quantum expectation value eq. (7.39) in terms of the diagrams of Fig. (7.1). For the expectation value of, for example $\langle\phi(x)\phi(y)\rangle$ we get the diagrams seen in the first line of Fig. (7.2). The first

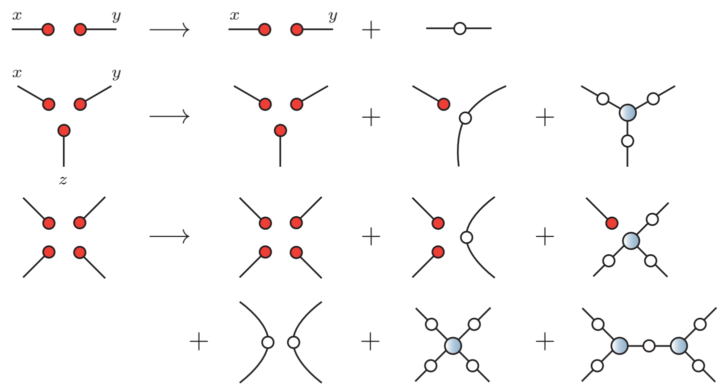


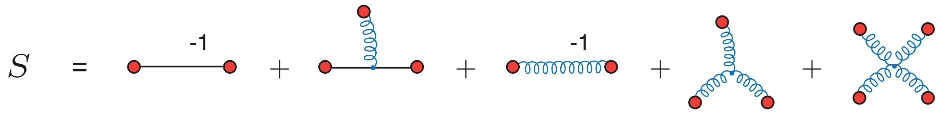
Figure 7.2: First line: graphical notation for expectation value $\langle\varphi(x)\varphi(y)\rangle$ using eq. (7.39). Second and third line: notation for higher n-point functions. (n=3 point function $\langle\phi(x)\phi(y)\phi(z)\rangle$, n=4 point function)

term after the arrow in the first line of Fig. (7.2) equals the first term in the brackets of eq. (7.39), which are the averaged fields and the classical expectation value. The second term equals the second term in brackets $\sim\int\Delta_{xy}i\delta/\delta\tilde{\phi}(y)$ which we can identify with the propagator between x and y with the sources removed, as seen in FIG.(7.1). In the end we set the sources to zero $J=0$ and in FIG.(7.2) all diagrams with source (red dots) vanish. Therefore only the propagator Δ_{xy} remains in our example. In other words the arrow in FIG.(7.2) shows going from classical theory $\rightarrow$ quantum theory, including quantum fluctuations.

In section (2.3) eq. (2.16) we derived a expression for the derivative of the effective action and since it is important for further proceeding let me quote it again

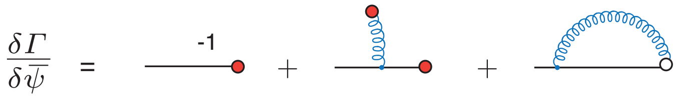
$$\Gamma_x[\tilde{\phi}]' = \frac{\delta}{\delta\phi}S\left[\tilde{\phi}(x) + \int_y \Delta_{xy}[\tilde{\phi}]\frac{\delta}{\delta\tilde{\phi}}\right]. \tag{7.40}$$

An this is all we need to derive the DSE, which will be done in four steps: **(i)** We start with the classical action from the QCD Lagrangian seen in eq. (2.8) . **(ii)** We calculate the classical equation of motion $\delta S/\delta\varphi$ as seen in eq. (7.30) and **(iii)** transfer this equation to the quantum equation of motion $\delta\Gamma/\delta\varphi$ using eq. (7.40). **(iv)** Last we take further derivatives of the effective action Γ'' , $\Gamma'''\ldots$.
For all the steps we can use the diagrammatic description we introduced above. Lets work this out for the QCD action
Step **(i)**:

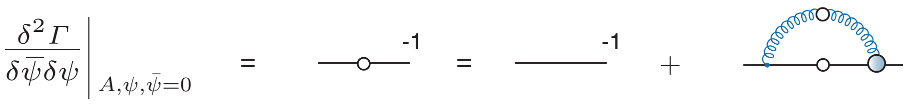


This is the graphical description of the classical QCD action, we should normally also include the ghost terms but we ignore them for this picture brevity and cause they are not important for the derivation of the quark DSE. Step **(ii)**

Taking the functional derivative with respect to the fermion field ψ. In the figure we can see that all diagrams that do not depend on the fermion field vanish. For Step **(iii)**



we use the powerful eq. (7.40) introduced above. In terms of the diagrams this means including all possible connections between the legs. This adds the third diagram to the picture. In the last Step **(iv)**



we take the second derivative of the generating functional with respect to the quark fields $\delta^2\Gamma/(\delta\bar{\Psi}\delta\Psi)$ and end up at the **Dyson-Schwinger equation** (DSE) for the quark. Taking the derivative removes the sources related to the fields.

After introducing the different functionals, deriving relation for functional derivatives and considering these in a diagrammatic description of the fields and propagators, we finally arrived at the DSE, the quantum equation of motion, within four steps. Similarly one can derive a DSE for the gluon or ghost propagator. For an easier example of φ^4-theory check [4].

7.3.3 Gauge-fixing working with the DSE's

For the functional approach (DSE) the Lagrangian must be gauge-fixed, which is provided by a gauge fixing term $\mathcal{L}_{g.f}$ we add to the Lagrangian. In Landau gauge this term follows as

$$\mathcal{L}_{g.f} = \frac{1}{2\alpha}\left(\partial_\mu A^i_\mu\right)^2 + \bar{c}^i\left(\partial^2\delta^{ij} + gf^{ijk}\partial_\mu A^k_\mu\right)c^j. \tag{7.41}$$

with the fields for the ghosts $\bar{c}^i$ c^j and the subscripts i, j, k denoting the color components, that are summed following Einsteins summation convention. By including the gauge-fixing term we fix the problem that the gluon propagator is not invertible. The expression in brackets $\bar{c}^i()c^j$ can be rewritten as $\partial_\mu D_\mu$ where here D_μ is the covariant derivative in the adjoint representation. The derivative in the adjoint representation means that the gluon field A_μ is in the adjoint representation, which means that the generator in $SU(3)$ equals the structure constant $(t^a)_{ij} = -if_{abc}$. The expression for the gauge-fixing term arises from evaluating the condition in the path integral. Through gauge-fixing we introduce these new unphysical fields, the ghosts. For a more detailed description on gauge-fixing check [20]. In this thesis I just want to emphasize that we have to gauge fix when working with the DSEs.

7.3.4 Hadron poles

The QCD Fock space contains all possible structures we think are contained in our world, like meson and bayrons as well as glueballs or even unphysical states(color states) and can be seen as a bigger version of the Hilbert space, since it also contains different n-particles states. All of these possible structures are eigenstates $|\lambda\rangle$ of the Hamiltonian, which carry momentum p and other quantum numbers like flavor, color etc. The completeness relation of the eigenstaten states in Fock space is given by

$$\mathbb{I} = \sum_\lambda \frac{1}{(2\pi)^3}\int \frac{d^3p}{2E_p}|\lambda\rangle\langle\lambda|, \tag{7.42}$$

where $E_p = p^2 + m^2_\lambda$, meaning that the momentum is on it's mass shell. Hadrons produce poles in QCD Green function and as one easy example for this I want to start with the two-point function, the propagator. Starting with the propagator I will insert an identity in form of the completeness relation and rewrite the fields $\phi(x)$ with translation operators. The fields ϕ are scalar fields in this case, but the derivation holds for all types of fields. First let's write the time ordering explicitly:

$$\langle 0|T\phi(x_1)\phi(x_2)|0\rangle = \Theta(x^0_1 - x^0_2)\langle 0|\phi(x_1)\phi(x_2)|0\rangle + \Theta(x^0_2 - x^0_1)\langle 0|\phi(x_2)\phi(x_1)|0\rangle \tag{7.43}$$

Now we take a look at the case $t_1 > t_2$, the derivation for the other case follows analog.

$$\begin{aligned}
\langle 0|\phi(x_1)\phi(x_2)|0\rangle &= \langle 0|\phi(x_1)\sum_\lambda \frac{1}{(2\pi)^3}\int \frac{d^3p}{2E_p}|\lambda\rangle\langle\lambda|\phi(x_2)|0\rangle \qquad (7.44)\\
&= \sum_\lambda \int \frac{d^3p}{(2\pi)^3 2E_p}\langle 0|e^{i\hat{H}t_1 - ipx_1}\phi(0)e^{-i\hat{H}t_1+ipx_1}|\lambda\rangle\\
&\times \langle\lambda|e^{i\hat{H}t_2 - i\hat{p}x_2}\phi(0)e^{-i\hat{H}t_2+i\hat{p}x_2}|0\rangle\\
&= \sum_\lambda \int \frac{d^3p}{(2\pi)^3 2E_p} e^{iE_{\lambda,\vec{p}}(t_1-t_2)+i\vec{p}\cdot(\vec{x_1}-\vec{x_2})}\langle 0|\phi(0)|\lambda_{\vec{p}}\rangle\langle\lambda_{\vec{p}}|\phi(0)|0\rangle\\
&= \int \frac{d^3p}{(2\pi)^3 2E_p} e^{i\omega_{\vec{p}}(t_1-t_2)+i\vec{p}\cdot(\vec{x_1}-\vec{x_2})}\langle 0|\phi(0)|\pi_{\vec{p}}\rangle\langle\pi_{\vec{p}}|\phi(0)|0\rangle + \sum_{\lambda\neq\pi}\cdots\\
&\underset{F.T}{=} \int \frac{d^3p}{(2\pi)^3 2E_p}\int d^4x\ e^{-ip'x}e^{i\omega_{\vec{p}}(t_1-t_2)+i\vec{p}\cdot(\vec{x_1}-\vec{x_2})}\cdot Z^2 + \sum_{\lambda\neq\pi}\cdots
\end{aligned}$$

where $Z^2 = \langle 0|\phi(0)|\pi_{\vec{p}}\rangle\langle\pi_{\vec{p}}|\phi(0)|0\rangle$ and only depends the four vector p^μ of the pion. Now we can use translation invariance to set the coordinates $x_1 = x$ and $x_2 = 0$ and we also evaluate the expression for different time ordering $t_2 > t_1$. Combining the two results we end up with

$$\langle 0|T\phi(x_1)\phi(x_2)|0\rangle = \int \frac{d^3p}{(2\pi)^3 2E_p}\int d^4x\ e^{-ip'x}e^{i\omega_{\vec{p}}|t|\pm i\vec{p}\cdot\vec{x}}\cdot Z^2 + \sum_{\lambda\neq\pi}\cdots .$$

For calculating the Fourier integral we will ignore the Z^2 factor for now. Note that for doing the integral a factor of $i\epsilon$ is needed in the exponential due to an equivalent reason we need it in the path integral. We stress here that the $i\epsilon$ is some fundamental to defining the field theory and not just something we put in last minute to save our integral, it has been there the whole time, even thought we did not explicitly write it out.

$$\begin{aligned}
\int d^3\vec{x}e^{-i\vec{p'}\cdot\vec{x}}\int_{-\infty}^{\infty} dt\ e^{ip'^0 t - iw_p|t|} &= (2\pi)^3\delta^3(\vec{p'}\pm\vec{p})\left(\int_{-\infty}^{0} dt\ e^{ip'^0 t + w_p t + i\epsilon} + \int_0^\infty dt\ e^{ip'^0 t - iw_p t - i\epsilon}\right) \qquad (7.45)\\
&= (2\pi)^3\delta^3(\vec{p'}\pm\vec{p})\left(\frac{1}{i(p'^0 + w_p)} - \frac{1}{i(p'^0 - w_p)}\right).
\end{aligned}$$

Putting everything together

$$= \int \frac{d^3p}{(2\pi)^3 2E_p}(2\pi)^3\delta^3(\vec{p'} \pm \vec{p})\left(\frac{1}{i(p'^0 + w_p)} - \frac{1}{i(p'^0 - w_p)}\right) \cdot Z^2 + \sum_{\lambda\neq\pi} \cdots \tag{7.46}$$

$$= \frac{1}{w_p}\left(\frac{1}{i(p'^0 + w_p)} - \frac{1}{i(p'^0 - w_p)}\right) \cdot Z^2 + \sum_{\lambda\neq\pi} \cdots \tag{7.47}$$

$$= \frac{Z^2}{i(p^2 - M_\pi^2)} + \sum_{\lambda\neq\pi} \cdots$$

with $p_0^2 - w_p^2 = p_0^2 - \vec{p}^2 - M^2 = p^2 - M^2$. We note that the final expression has the expected pole contribution and the amplitudes Z^2 projecting onto the pion state. Working this out for the four-point functions follows accordingly, the completness realtion is inserted in the middle and we will end up with expression for Z^2 that contain two field sandwiched in the vacuum state and the hadron state.

7.3.5 Renormalization and Regularization

For the purpose of defining the theory we do two basics steps

- **Regularization**
- **Renormalization**.

To get started I want to emphasize the mass-dimensions of the coupling constant in our theory. This will help us understand which theory is renormalizabel and which is not. In quantum field theory it is standard to work in units for which $\hbar = c = 1$. In this unit system every quantity has units equal to some power of energy. It is also standard to use shorthand $[X] = n$ or $dX = n$ if the quantity has units equal to E^n. The dimension of the action is zero $[S] = 0$, which implies that the Lagrangian has the mass-dimension equal the dimension of space tine d, $[L] = d$. In a scalar field theory the coupling constant has dimensions $[g] = d + p - p \cdot d/2$, with p being the exponent of the field (in a φ^4-theory $p = 4$). A amputated n-point function $\Gamma^n(x_1 \dots x_n)$ has mass dimensions $[\Gamma] = d + n - nd/2$. In four dimension this would account to $[g] = 4 - p$ and $[\Gamma] = 4 - n$.

We can also guess the mass dimension by analyzing Feynman diagrams, counting the number of loops L. Each loop comes with a mass dimension of d. The number of propagators I in a diagram with mass dimension -2 and the number of vertices V with mass dimension of $[g]$. We define a parameter D as

$$D = Ld - 2I. \tag{7.48}$$

D is the so called the superficial degree of divergence. This parameter describes the divergence of a diagram. Each closed loop gives a factor of d^dl and each internal propagator a factor of $1/(k^2+m^2)$, where k^2 is some linear combination of the loop momentum

l and the external momentum. A integral, or a diagram equivalently, will be divergent if the loop integration overtakes the momentum power in the denominator. D can also be defined in terms of the parameters mentioned above:

$$d_\Gamma = dL - 2I + [g]V = D + [g]V \quad \Rightarrow \quad D = d_\Gamma - [g]V. \tag{7.49}$$

$D \geq 0$ means the diagram is divergent, $D < 0$ convergent. But it is superficial, so a diagram might diverge even if $D < 0$ and the opposite way around.
Lets define what we understand under a "**Renormalizable theory**":
A theory is renormalizable if the mass dimension of its coupling constant is greater equal zero :

$$[g] \geq 0 \tag{7.50}$$

or different: *a theory is non-renormalizable if any coefficient of any term in the Lagrangian has none negative mass dimensions*

This is also indicated by the fact that there is just a finite number of renormalization constant needed to make the theory finite. In a nutshell renormalization means we can get rid of unpleasant infinities if only a finite number of counter terms or conditions need to be introduced to make the theory finite.
φ^4 is an example for a renormalizable theory since $[g] = d - P\frac{d-2}{2} = 0$ for $d = 4$ and $p = 4$. Also QCD is a renormalizable theory, as treated so far, since the theory is described in $d = 4$ dimensions and $[\Psi A \Psi] = 4 \Rightarrow\ [g] = 0$. In case of vector particles gauge is only needed to verify the argument of renormalizability of QCD.

Some short remarks about **regularization**.

There are different ways to do regularize a theory, like dimensional regularization, Pauli-Villar or Lattice-gauge theory, where the lattice spacing a is introduced as the cut off scale. Once we use a regulator to identify the infinities we want to get rid off them and we achieve this by enforcing three **renormalization conditions**.

For a scalar field theory we want to enforce the condition that the propagator has a certain value at the renormalization point μ, as

$$\begin{aligned} &\textbf{1.)}\ \ \Delta^{-1}(p)|_{p^2=\mu^2} \stackrel{!}{=} p^2 - m^2 \\ \textit{or}\quad &\frac{d}{dp^2}\Delta^{-1}(p)|_{p^2=\mu^2} \stackrel{!}{=} 1\ . \end{aligned} \tag{7.51}$$

This fixes two of the three renormalization conditions, the last condition is for a generic vertex function and we fix

$$\textbf{2.)}\ \ \Gamma(\{p_i\})|_{p^2=\mu^2} \stackrel{!}{=} g \tag{7.52}$$

This scheme is called the **MOM-scheme** (momentum subtraction scheme) or Mini-Mom scheme. Setting the three conditions means we set global conditions on our Green functions. The most known scheme is the MS or $\bar{M}S$ scheme. The difference to the MOM-scheme is, that instead of applaying three conditions on the propagator and vertices, counter-terms are introduced in the Lagrangian to cancel the infinities. Implicitly there are also counter terms introduced when we are using the MOM-scheme, but by enforce the renormalization conditions we are choosing this counter terms without worrying how exactly they look like.

Rather than working with counter terms renormalization constants for are introduced and chosen to fulfill the renormalization conditions. In course of this thesis and working with the DSEs a similar scheme called the minimal MOM scheme (Mini-MOM) is used. A explicit perturbative discussion of the scheme and mathematical comparison to the $\bar{M}S$ scheme can be found in [39].

Renormalization in QCD As mentioned in the beginning of section (7.3.5), QCD is a renormalizable theory because its coupling constant g is mass dimensionless and because if gauge invariance. If we take a look at the QCD action in eq. (2.8) and add the gauge-fixing term that was introduced later, eq. (7.41), we note that there are several fields that need renormalization constants, alias counter terms and we relate the field to their renormalized quantities as:

$$\psi_B = Z_2^{1/2}\psi, \quad A_B = Z_3^{1/3}A, \quad c_B = \tilde{Z}_3^{1/2}c, \quad m_B = Z_m m, \quad g_B = Z_g g. \tag{7.53}$$

We start with giving each piece in the Lagrangian its own renomalization constant, for example introducing Z_{1F} for the quark-gluon vertex or $\tilde{Z}_1$ for the ghost-gluon vertex. Doing so we would end up with

$$\begin{aligned}
\mathcal{L}_{QCD} &= Z_2\bar{\psi}(i\partial\!\!\!/ - Z_m M)\psi + Z_{1F}g\bar{\psi}A\!\!\!/\psi + Z_3\frac{1}{2}A^a_\mu(\Box g_{\mu\nu} - \partial_\mu\partial_\nu)A^a_\nu && (7.54)\\
&- Z_1\frac{g}{2}f_{abc}(\partial_\mu A^a_\nu\partial_\nu A^a_\mu)A^b_\mu A^c_\nu - Z_4\frac{g^2}{4}f_{abe}f_{cde}A^a_\mu A^b_\nu A^c_\mu A^d_\nu && (7.55)\\
&+ Z_\chi\frac{A^a_\mu\partial_\mu\partial_\nu A^a_\nu}{2\chi} + \tilde{Z}_3\bar{c}a\Box c_a + \tilde{Z}_1 ig(\partial_\mu\bar{c}_a)[A_\mu, c]_a. && (7.56)
\end{aligned}$$

with the $\Box$ denoting the d'Alembert operator, the field are given according to the action eq .(2.8) and f_{abc} are the structure constants. As a consequence of gauge invariance the are relation between these renormalization constants, which means we do not have to calculate all of them. These relations come from the Slavonov-Taylor identities and read

$$Z_{1F} = Z_g Z_2 Z_3^{1/2}, \quad Z_1 = Z_g Z_3^{3/2}, \quad Z_4 = Zg^2 Z_3^2, \quad \tilde{Z}_1 = Z_g Z_3^{1/2}\tilde{Z}_3. \tag{7.57}$$

7.4 The Bethe-Salpeter wavefunction

The **Bethe-Salpeter Wave-function** χ (BSW) is the non-amputated version of the Bethe-Salpeter amplitude Γ and is defined through $\chi = S_+ \Gamma S_-$, where $S_\pm$ is the quark propagator at $q_+ = q + \sigma P$ and $q_- = q - (1-\sigma)P$ with the momentum distribution parameter $\sigma \in \{0, 1\}$. The BSW χ has the same quantum number as the BSA and can thus be expressed through the basis functions τ_i introduced in section(3.2). Using this relation we can rewrite the BSW as follows

$$\begin{aligned}
\chi(q,P) &= S(q_+)\Gamma(q,P)S(q_-) \qquad (7.58)\\
&= \sum_{i=0}^{3} \tau_i(q,P)\chi_i(q,p)\\
&= \sum_{i=0}^{3} S(q_+)\tau_i(q,P)S(q_-)\Gamma_i(q,P).
\end{aligned}$$

Here Γ_i should indicate the dressing functions of the BSA and χ_i the dressing functions of the BSW. Applying projectors $P_i(q,P)$ on both sides of eq. (7.58) gives

$$\chi_i(q,P) = \sum_{j=0}^{3} Tr\left[P_i(q,P)S(q_+)\tau_j(q,P)S(q_-)\right]\Gamma_j(q,P) \qquad (7.59)$$

where the trace over the expression in rectangular brackets is the BSA kernel matrix or **rotation matrix** $Y_{ij} = Tr[\dots]$. The kernel Y_{ij} can be seen as a rotation in the space of Γ_i, since both the BSA and the BSW live in the same space and are connected through $\chi_i = Y_{ij}\Gamma_j$. If we insert the expression for χ eq. (7.58) into the expression for the BSA eq. (3.35), derived in section(3.2) and project again with $P_i(q,P)$ on both sides we obtain

$$\begin{aligned}
\Gamma_i(q,P) &= -\frac{4}{3}\int d\bar{k} \left(D_{\mu\nu} \sum_{j=0}^{3} Tr\left[P_i(q,P)\gamma_\mu T_j(k,P)\gamma_\nu\right]\right)\chi_i(k,P) \qquad (7.60)\\
&= -\frac{4}{3}\int d\bar{k} \left(\sum_{j=0}^{3} L_{ij}\right)\sum_{k=0}^{3} Y_{jk}(k,P)\Gamma_k(k,P) \qquad (7.61)
\end{aligned}$$

Since the **BSW** χ_i is a scalar dressing function we can extract it from the trace. The final equation eq. (7.61) includes two summations running over the four basis structures $(j, k = 0 \dots 3)$ and two traces.

7.5 quark-photon-vertex parameterization

The model for the quark-photon vertex used in this thesis is a simple combination of polynoms times exponential, achieved through fitting to the calculated dressing functions f_i in [25]. The parameterization reads as

$$f_i = \left(p_{11} + p_{21}S_0 + p_{31}S_0^2\right) e^{-p_{41}S_0} + S_0 s \left(p_{12} + p_{22}S_0 + p_{32}S_0^2\right) e^{-p_{42}S_0} \tag{7.62}$$

with the fittet parameter in the table on the next page.

f_i, i	j	p_{1j}	p_{2_j}	p_{3j}	p_{4j}
1	1	-2.2931561	3.0883036	-2.8005863	2.5011909
1	2	-4.8714375	8.2453213	-0.0709300	6.3633555
2	1	1.0872236	-1.6937272	2.1697800	5.6191265
2	2	0.6841379	-2.9447699	1.9623880	3.9297677
3	1	-0.0148812	0.4368582	0.0002088	5.3150391
3	2	0.8951350	-1.4679810	0.9870903	3.9391567
4	1	2.2316731	-3.2576626	3.0619956	2.4460796
4	2	3.3830657	-8.8023358	-0.0430093	6.2426821
5	1	-4.4056266	4.6579152	-6.4524247	4.5011318
5	2	-9.4101894	22.6974443	-0.1624843	6.9495643
6	1	-1.4822436	5.6963474	0.0037994	5.4008074
6	2	0.3814822	-0.5617195	0.1658344	0.9363245
7	1	2.1109834	-3.0618272	3.8296863	3.9577049
7	2	2.2861203	-10.1585214	-0.0106440	6.4476713
8	1	-1.1267876	2.1122487	-2.8972814	5.0593727
8	2	-2.2809668	8.1152154	0.0513090	8.2730966

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Erklärung

Hiermit erkläre ich an Eides statt, dass ich die vorliegende Master-Thesis selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe.

Gießen, den February 17, 2016

Esther Weil

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